Bauhaus-Universität Weimar
Faculty of Media
Study Programme Computer Science for Digital Media

Reproducing and Evaluating State-of-the-Art Methods In Chest X-ray Abnormalities Detection

Master's Thesis

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Submission date: October 6, 2025

Declaration of Authorship

I, **Sahand Majdabadi Farahani**, declare that this thesis titled, "*Reproducing and Evaluating State-of-the-Art Methods in Chest X-ray Abnormalities Detection*," And the work presented in it is my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed: _____

Date: 6.10.2025

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Abstract

Faculty of Media
Chair of Computer Vision in Engineering

Master of Science

Reproducing and Evaluating State-of-the-Art Methods in Chest X-ray Abnormalities Detection

by Sahand MajdabadI Farahani

The prevalence of digital radiography and the scale of public datasets have accelerated research on automatic detection of thoracic abnormalities. This thesis reproduces competitive Kaggle solutions for the VinBigData challenge and evaluates modern detectors across two phases: (i) CNN-based models (YOLOv5, EfficientDet-D2, Faster R-CNN, VFNet) and (ii) transformer detectors (DETR R50 and Co-DETR/Co-DINO with Swin-L). A unified benchmarking pipeline is built with COCO-format data, careful preprocessing (weighted box fusion for label consolidation), and stratified cross-validation. Results show that reproduced CNN baselines closely approach official leaderboard scores, while transformer models provide competitive accuracy with simpler, end-to-end training. We systematically analyze trade-offs between speed, localization quality, and robustness to class imbalance. The complete code, configuration files, and experiment logs are released for transparency and reuse at `https://github.com/SahandMF/vinbig-reproduction`.

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List of Abbreviations

AI	Artificial Intelligence
AP	Average Precision
AP50	Average Precision at IoU threshold of 0.50
BiFPN	Bidirectional Feature Pyramid Network
CNN	Convolutional Neural Network
COCO	Common Objects in Context (dataset and evaluation standard)
CV	Cross-Validation
CXR	Chest X-ray
DETR	DEtection TRansformer
DICOM	Digital Imaging and Communications in Medicine
EMA	Exponential Moving Average
FPN	Feature Pyramid Network
GPU	Graphics Processing Unit
IoU	Intersection over Union
JSON	JavaScript Object Notation
LB	Leaderboard (Kaggle competition ranking)
MAE	Masked Autoencoder
mAP	mean Average Precision
ML	Machine Learning
MMDetection	OpenMMLab Object Detection Framework
MMYOLO	OpenMMLab YOLO-based Object Detection Framework
NMS	Non-Maximum Suppression
PNG	Portable Network Graphics
ReLU	Rectified Linear Unit
RPN	Region Proposal Network
SOTA	State-of-the-Art
TTA	Test-Time Augmentation
VFNet	VarifocalNet
ViT	Vision Transformer

WBF	Weighted Boxes Fusion
YOLO	You Only Look Once (object detection family)
WandB	Weights & Biases (experiment tracking platform)

Chapter 1

Introduction

Chest radiography remains a central diagnostic tool in everyday clinical practice. Yet the growing volume of examinations, the subtle presentation of many thoracic findings, and variability in image quality make consistent interpretation difficult. In parallel, modern computer vision has produced a diverse set of object–detection methods that can localize disease patterns directly in an image. This thesis examines how such methods perform on a large, publicly available benchmark for chest X–ray abnormality detection, with a focus on reproducibility and fair comparison. The aim is twofold: to faithfully reconstruct strong competition baselines and to evaluate recent architectural advances under a unified, transparent pipeline suitable for medical imaging research.

1.1 Background and Motivation

Chest X-rays (CXRs) are among the most common and cost-effective tools used in hospitals to examine the chest area. They help doctors identify various medical problems such as pneumonia, lung nodules, or heart-related issues. However, reading and interpreting these images can be difficult, even for trained radiologists. Problems such as low image contrast or overlapping body structures can make it hard to detect all signs of disease. In busy hospitals or in regions with few specialists, this problem becomes even more serious.

Because of this, there is growing interest in using artificial intelligence (AI) to support medical image diagnosis. Deep learning models—especially those used for object detection—can help locate and label abnormal areas in chest X-rays. They can save time for radiologists, improve diagnostic accuracy, and make sure that fewer cases are missed. These systems are already being tested in hospitals and continue to show promising results [1].

One important benchmark for this research is the VinBigData Chest X-ray Abnormalities Detection challenge, which took place on Kaggle in 2021. See Fig. 1.1 [2].

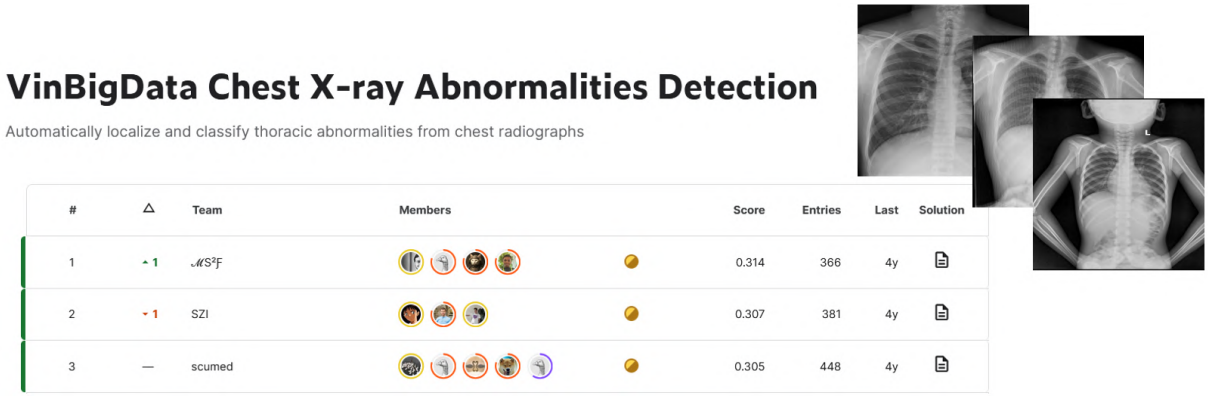


Figure 1.1: An overview of Kaggle's VinBigData Chest X-ray Abnormalities Detection The challenge included the final winners group on the leaderboard and some dataset samples.

In this competition, participants worked with a real-world dataset of CXRs, labeled by professional radiologists. The goal was to detect up to 14 different abnormalities per image, which made it a challenging multi-label object detection task.

Table 1.1 presents three sample images for each of the 14 abnormality classes in the VinBigData chest X-ray dataset. The samples show the variety of visual patterns that make this detection task especially complex for machine learning models. All images were manually selected to illustrate visible and diverse presentations of each condition.

Table 1.1: Sample images for each class label from the VinBigData chest X-ray dataset.

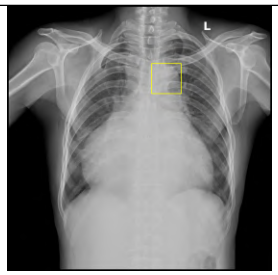
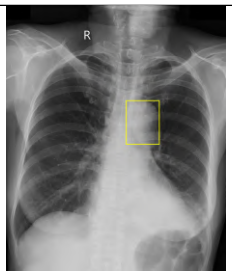
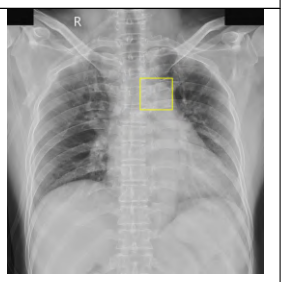
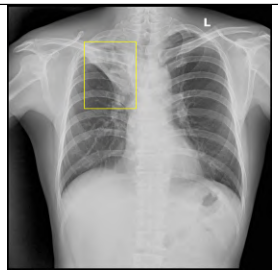
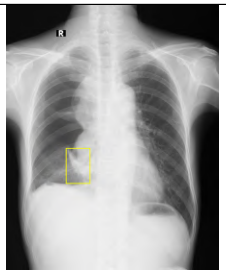
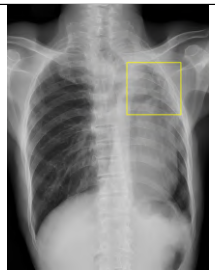
Class Label	Sample 1	Sample 2	Sample 3
Aortic enlargement			
Atelectasis			

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Table 1.1 – continued from previous page.

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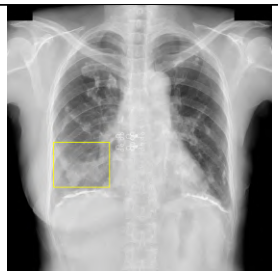
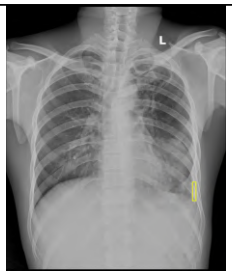
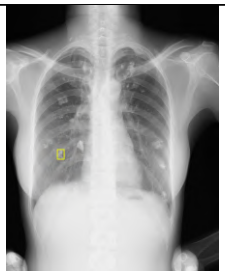
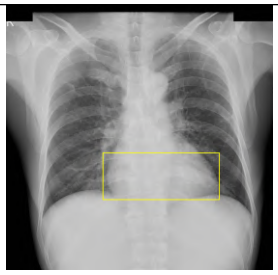
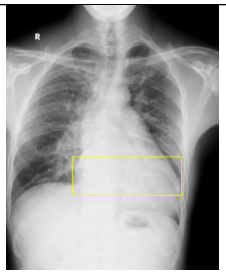
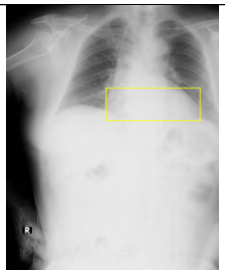
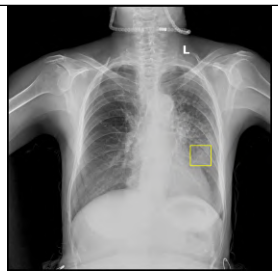
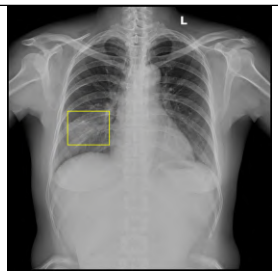
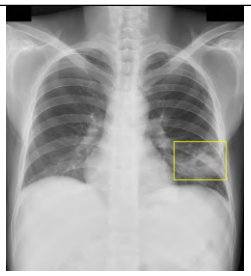
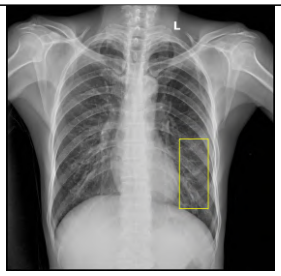
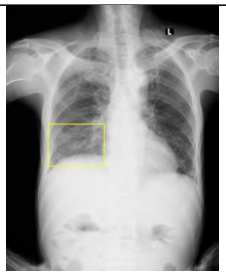
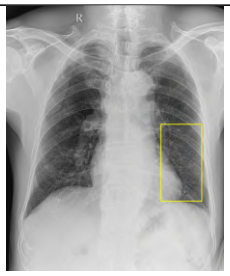
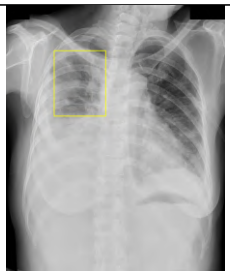
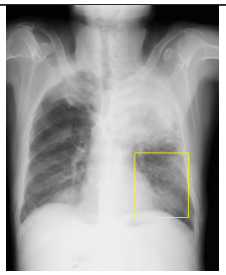
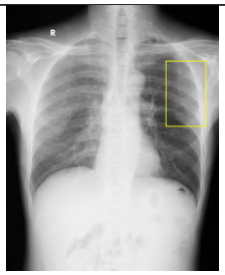
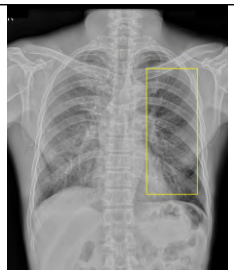
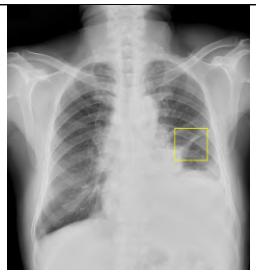
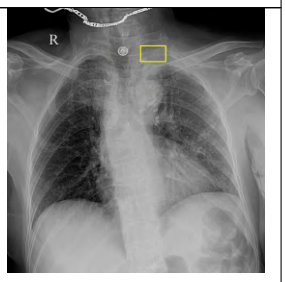
Calcification			
Cardiomegaly			
Consolidation			
ILD			
Infiltration			
Lung Opacity			

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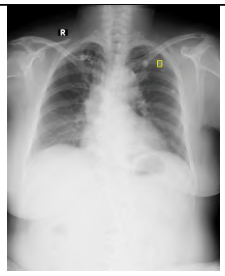
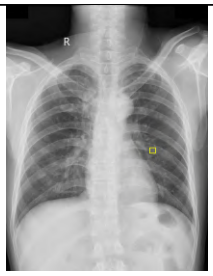
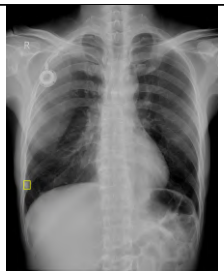
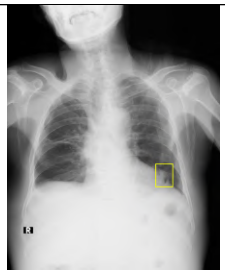
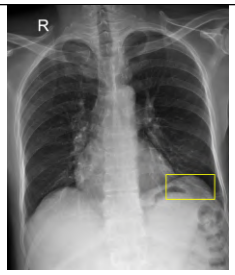
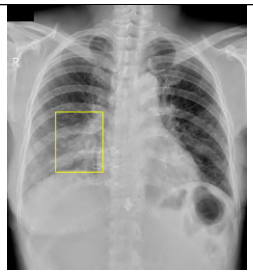
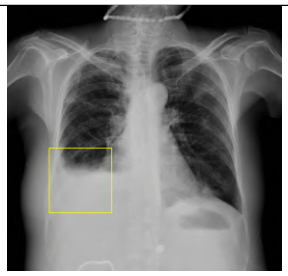
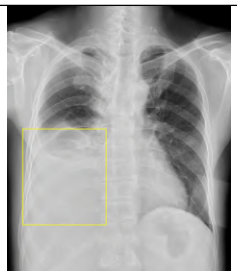
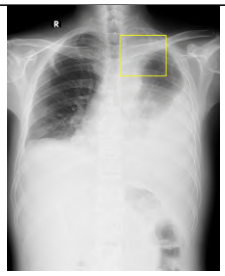
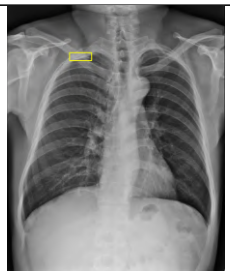
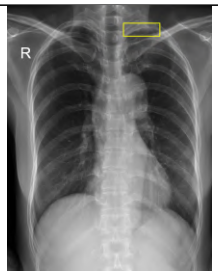
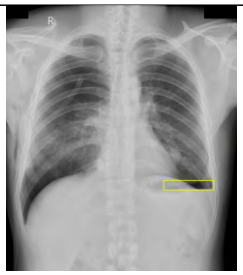
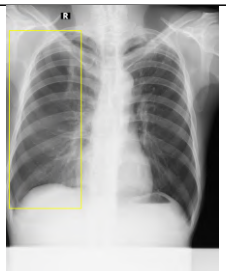
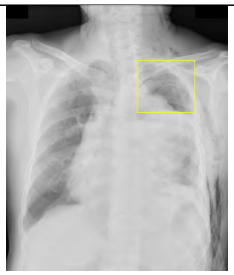
Nodule/Mass			
Other lesion			
Pleural effusion			
Pleural thickening			
Pneumo-thorax			

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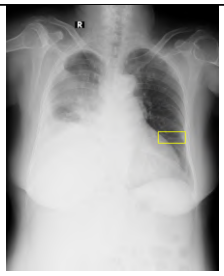
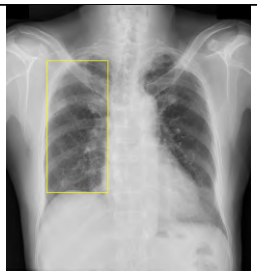
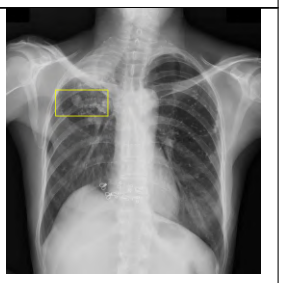
Table 1.1 – continued from previous page.

Class Label	**Sample 1**	**Sample 2**	**Sample 3**
Pulmonary fibrosis			

The top solutions used strong models such as YOLOv5 [3], EfficientDet [4], VFNet [5], and Faster R-CNN [6]. YOLOv5 follows the one-stage, dense-prediction "You Only Look Once" design introduced by Redmon et al. in YOLOv1 [7] and extended in YOLO9000/ YOLOv2 [8] and YOLOv3 [9], prioritizing speed by predicting bounding boxes and class probabilities directly from a single forward pass over the image. EfficientDet integrates an EfficientNet backbone with a BiFPN and employs compound scaling to jointly adjust network depth, width, and resolution for a balanced trade-off between accuracy and efficiency. VFNet incorporates IoU-aware classification through Varifocal Loss, aligning classification confidence with localization quality to improve ranking of high-quality detections. Faster R-CNN, a representative two-stage detector, uses a Region Proposal Network (RPN) to generate candidate regions, which are then refined by a second-stage classifier and bounding box regressor. Since the competition, transformer-based detectors such as DETR [10] have reformulated object detection as a set prediction problem with bipartite matching, while improved loss functions like Generalized IoU [11] and training strategies such as curriculum learning [12] have also gained traction. Ensembling methods such as Weighted Boxes Fusion (WBF) [13], [14] are frequently applied to combine predictions from multiple models, often yielding incremental improvements in mean Average Precision (mAP).

Another major motivation for this thesis is reproducibility. In machine learning, especially in applied fields like medical imaging, it is often difficult to reproduce the exact results from published papers or competitions [15]. Missing code, unclear settings, or data processing errors can make results unreliable. In a field like healthcare, where decisions affect human lives, this lack of transparency is a serious problem.

This thesis wants to address both of these points: First, by recreating and comparing top competition solutions in open-source frameworks like MMDetection [16] and MMYOLO [17]. Second, by testing new ideas and showing how they perform on the same dataset. The goal is not only to reach strong performance but also to make the results easy to understand and repeat by others.

1.2 Research Problem

Despite recent advances in deep learning for medical imaging, applying these methods to real-world datasets still brings several practical challenges. One major issue is the *gap between competitive results and real reproducibility.* While top solutions in challenges like the VinBigData Chest X-ray Abnormalities Detection competition achieve high accuracy, they are often hard to fully replicate. The reasons include missing training details, unavailable code, undocumented preprocessing steps, or dependence on specific hardware.

This reproducibility gap makes it difficult for researchers to build on past work or confidently evaluate new methods. It also limits the trust that clinicians and hospitals can place in these models, especially in high-stakes medical applications.

In addition, most published work focuses on individual model performance — for example, how well EfficientDet or YOLOv5 performs on its own. But in practice, multiple design choices affect model success:

- Which **framework** is used (e.g., MMDetection vs. MMYOLO)?
- What **augmentation** or **normalization** strategy is applied?
- Are **ensembling** methods like Weighted Boxes Fusion (WBF) used [13]?

These design choices are rarely studied side-by-side using consistent setups. As a result, it's unclear how different models really compare when evaluated fairly on the same dataset, with the same pipeline.

At the same time, **new architectures** like transformers (e.g., DETR [10]) and improved loss functions (e.g., Generalized IoU [11]) have emerged since the original VinBigData challenge. But whether these newer techniques *outperform* the older baseline solutions — or under what conditions — is still an open question.

This thesis addresses the following core research questions:

- **RQ1:** Can we fully reproduce top solutions from the VinBigData competition using modern open-source libraries?
- **RQ2:** How do state-of-the-art methods perform under a unified benchmarking setup?
- **RQ3:** What impact do architectural changes, training strategies, and ensembling methods have on detection performance?

By answering these questions, the goal is not only to improve accuracy but also to promote transparency and reproducibility in medical AI research.

1.3 Objectives of the Thesis

The primary goal of this thesis is to systematically reproduce, benchmark, and extend state-of-the-art object detection models for chest X-ray abnormality detection using the VinBigData dataset. The overall workflow is divided into two major phases: (1) reproduction of top competition solutions as a proof of concept, and (2) exploration of modern improvements in model architecture, training strategies, and ensembling.

Phase 1: Reproduction and Proof of Concept

This phase focuses on replicating the key components and performance levels of the most successful submissions in the VinBigData Kaggle competition. Specifically, the 1st, 4th, and 10th place solutions are targeted for reproduction using modern open-source frameworks such as MMDetection, MMYOLO, and EfficientDet-PyTorch. Each of these solutions is based on well-known object detection models, including YOLOv5, EfficientDet-D2, VFNet, and Faster R-CNN.

The main objectives in this phase are:

- Reconstruct the original training and evaluation pipeline for each of the selected solutions.
- Implement the models in standardized, modular frameworks with minimal custom code.
- Validate that each model achieves similar performance to the originally reported results when retrained from scratch.
- Ensure that all components (data loading, preprocessing, augmentation, loss functions, and metrics) are transparent and reproducible.

This phase serves both as a functional verification step and as a solid baseline for the improvements tested in Phase 2.

Phase 2: Exploration of Modern Improvements

After the reproduction of baseline models, the second phase investigates recent advances in object detection and evaluates their effectiveness on the same dataset. These include:

- **New Architectures:** Transformer-based detectors such as DETR [10] are tested and compared with traditional CNN-based models.
- **Advanced Data Augmentation:** The impact of different augmentation schemes and normalization techniques is systematically studied.
- **Modern Ensembling Techniques:** Post-processing strategies like WBF [13] are applied to combine model outputs and improve performance.

Each of these techniques is evaluated under the same controlled setup as the baseline models to ensure fair comparison. The focus is not only on improving performance but also on identifying which innovations actually contribute to robustness and generalization in medical object detection tasks.

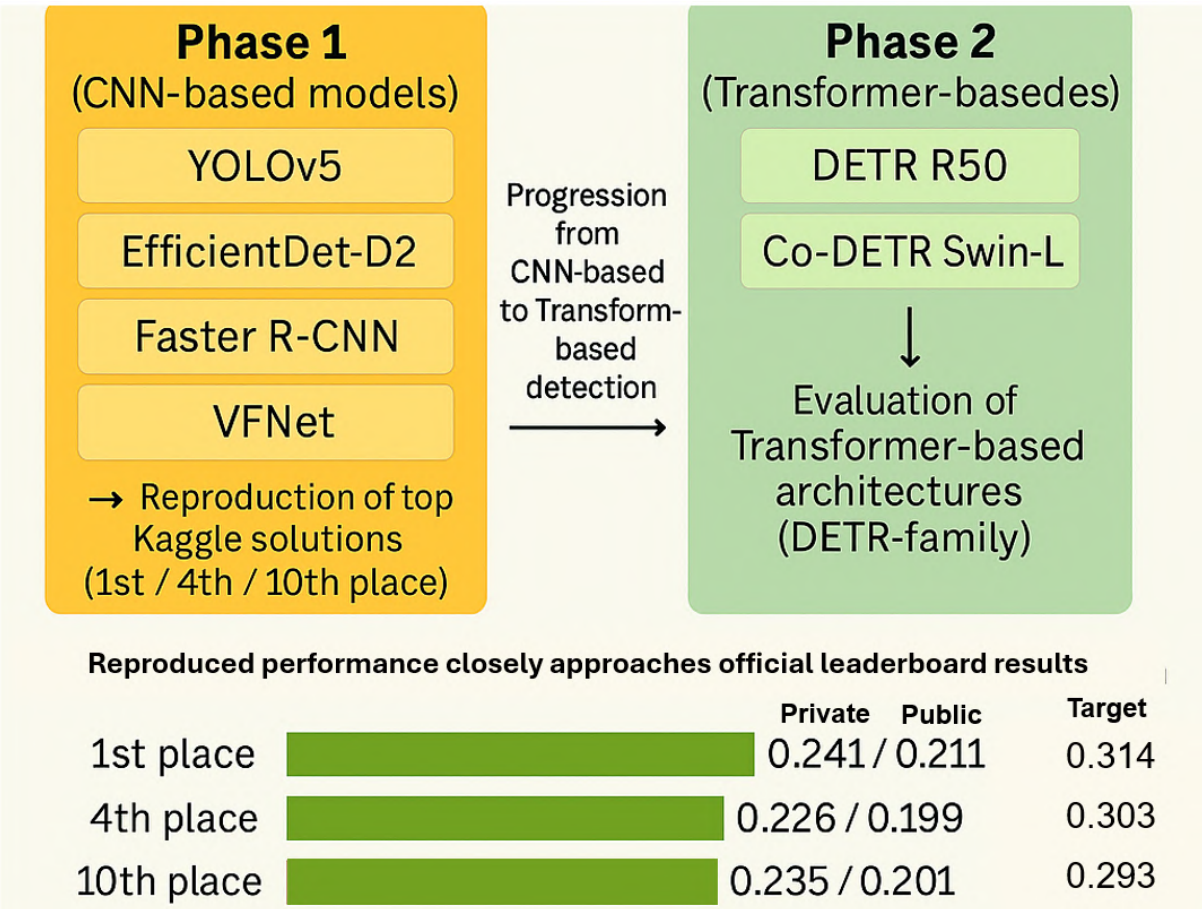


Figure 1.2: Overview of the two-phase experimental design. **Phase 1** focuses on reproducing top-performing CNN-based Kaggle solutions (YOLOv5, EfficientDet-D2, Faster R-CNN, and VFNet), while **Phase 2** evaluates transformer-based detectors (DETR R50 and Co-DETR Swin-L). The reproduced results closely match the official Kaggle leaderboard scores, confirming the reliability and reproducibility of the proposed benchmarking framework.

Summary of Research Objectives

To summarize, the thesis is structured around the following concrete objectives:

- **O1:** Reproduce and validate the top-performing solutions from the VinBigData competition using open-source tools.
- **O2:** Establish transparent, modular, and fully reproducible baselines for each model and framework.
- **O3:** Explore the impact of recent architectural innovations and training techniques on detection accuracy.
- **O4:** Benchmark different model components (architecture, augmentation, ensembling) in a unified experimental pipeline.
- **O5:** Deliver a clean, well-documented codebase that can be reused for further research or clinical prototyping.

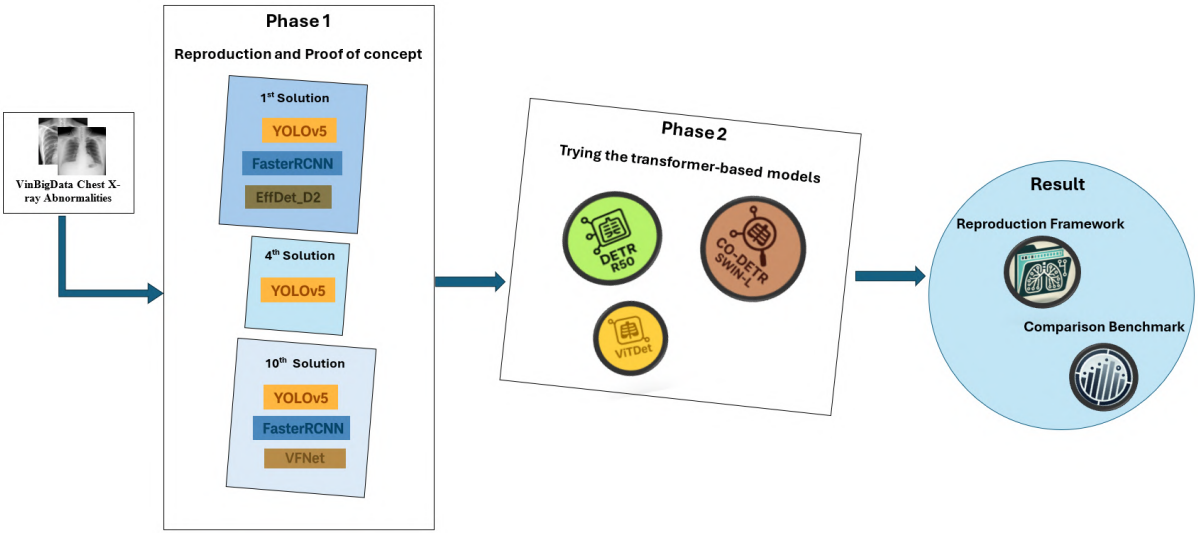


Figure 1.3: Roadmap of the thesis: starting with reproduction of competition solutions as proof of concept, followed by exploration of new model architectures.

Chapter 2

Related Work

This chapter provides an overview of the research foundations underlying the models and design choices evaluated in this thesis. A brief introduction to object detection is first presented, outlining the backbone–neck–head architecture and the one– versus two–stage taxonomy that characterizes most modern detectors. The core convolutional architectures employed in the experiments—Faster R–CNN, YOLOv5, EfficientDet, and VFNet—are then reviewed, emphasizing their feature extraction mechanisms, multi-scale fusion strategies, and prediction heads. Subsequently, transformer-based detectors are examined, with particular focus on DETR and its successors, which reformulate object detection as a set prediction problem. As training procedures can strongly influence model performance, data augmentation techniques that enhance generalization are also summarized. The chapter concludes with a discussion on reproducibility, which forms the basis for the standardized benchmarking framework applied throughout this study.

2.1 Object Detection Basics

Object detection is a central task in computer vision that combines two goals: identifying the category of objects in an image and localizing them with bounding boxes. In contrast to image classification, which assigns one or more labels to an entire image, object detection must also determine *where* in the image each object is located.

Most modern detectors share a common three-part pipeline: **backbone**, **neck**, and **head**. Each of these components plays a distinct role in transforming an input image into final predictions.

Backbone. The backbone is the feature extractor. Typically, it is a convolutional neural network such as ResNet [18] or EfficientNet [19] that transforms the raw image into feature maps. These maps capture hierarchical information, from low-level edges and textures to high-level semantic patterns. Because objects vary in scale, the backbone usually outputs multiple levels of features, which are later combined by subsequent modules.

Neck. The neck is an intermediate module designed to enhance and organize the backbone's features before they reach the detection head. The most common designs are Feature Pyramid Networks (FPN) [20] and their variants, such as Bidirectional Feature Pyramid Network (BiFPN) [4]. These architectures combine feature maps from different resolutions, allowing the detector to recognize both small and large objects more reliably. In practice, the neck acts as a bridge between the raw feature extraction of the backbone and the task-specific predictions of the head (Fig. 2.1).

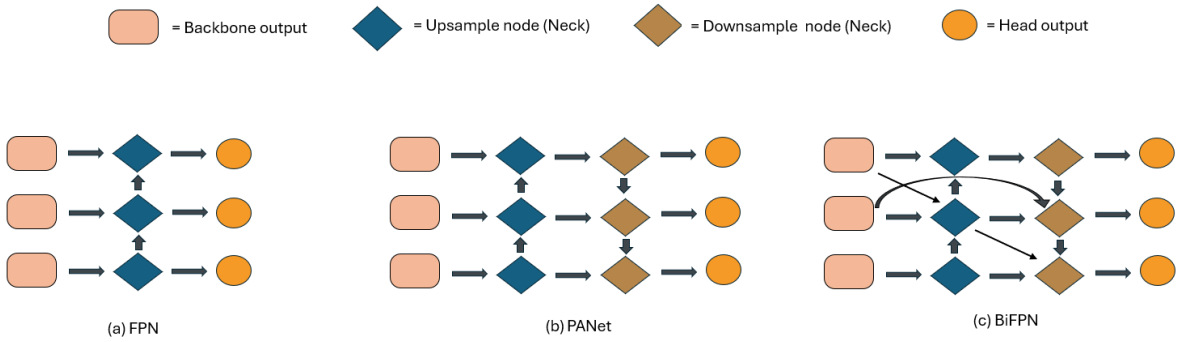


Figure 2.1: Comparison of feature fusion architectures used as necks in object detectors. (a) FPN uses a top-down pathway with lateral connections. (b) PANet enhances this with an additional bottom-up path for improved information flow. (c) BiFPN introduces weighted bidirectional connections to optimize multi-scale feature fusion further. Figure reproduced and adapted for illustration based on [4], [21].

Head. The head is the prediction module. It takes the processed features from the neck (or directly from the backbone in simpler models) and outputs bounding boxes and class probabilities. Different detection paradigms arise depending on how the head is designed:

- In **two-stage detectors** such as Faster R-CNN [6], the head operates in two steps. An RPN first generates candidate object regions, and a second classifier and regressor refine these proposals. This separation typically improves accuracy, but requires more computation.

- In **one-stage detectors** such as YOLOv5 [3] or EfficientDet [4], the head directly predicts bounding boxes and classes from feature maps without a proposal stage. This design is faster and more efficient, making it well suited for real-time or resource-constrained applications.

More recently, a third line of approaches has emerged with **transformer-based detectors**. Instead of using anchors or dense proposals, they treat detection as a *set prediction* problem and decode object hypotheses directly with attention mechanisms. This direction will be discussed in Section 2.3.

Overall, the backbone–neck–head framework provides a useful taxonomy for analyzing detection models. Backbones focus on feature extraction, necks aggregate features across scales, and heads specialize them for classification and localization. This modular view also clarifies the differences between the architectures explored in this thesis, which are presented in Section 2.2.

2.2 Core Architectures

The top-performing solutions in the VinBigData Chest X-ray Abnormalities Detection Challenge were based on state-of-the-art object detection architectures, each representing a distinct design philosophy. These models include YOLOv5 [3], EfficientDet [4], VFNet [5], and Faster R-CNN [6], which were frequently complemented by advanced augmentation and ensembling strategies to maximize performance.

Faster R-CNN

Faster R-CNN [6] is one of the most influential two-stage object detection frameworks. Its design separates the tasks of region generation and object classification, which helps to improve both localization and detection quality. A convolutional backbone, such as ResNet [18], is first used to extract feature maps from the input image. On top of these maps, a RPN generates candidate regions that are likely to contain objects. This proposal step replaces earlier methods such as Selective Search [22], making the entire pipeline faster and trainable end-to-end.

In the second stage, the proposed regions are refined and classified. Features corresponding to each candidate region are pooled from the backbone output and passed to a detection head, which outputs a class label and a more accurate bounding box. By decoupling region generation from classification, Faster R-CNN typically achieves stronger localization accuracy than one-stage detectors, particularly for irregularly shaped or subtle objects. The trade-off, however, is higher computational cost and slower inference speed compared to one-stage approaches.

Overall, Faster R-CNN established the modern two-stage paradigm and demonstrated that region proposals integrated with deep feature extraction can significantly improve detection performance. Its influence extends to many subsequent models, and it remains a widely used baseline in both general-purpose vision tasks and specialized applications such as medical image analysis(Fig. 2.2).

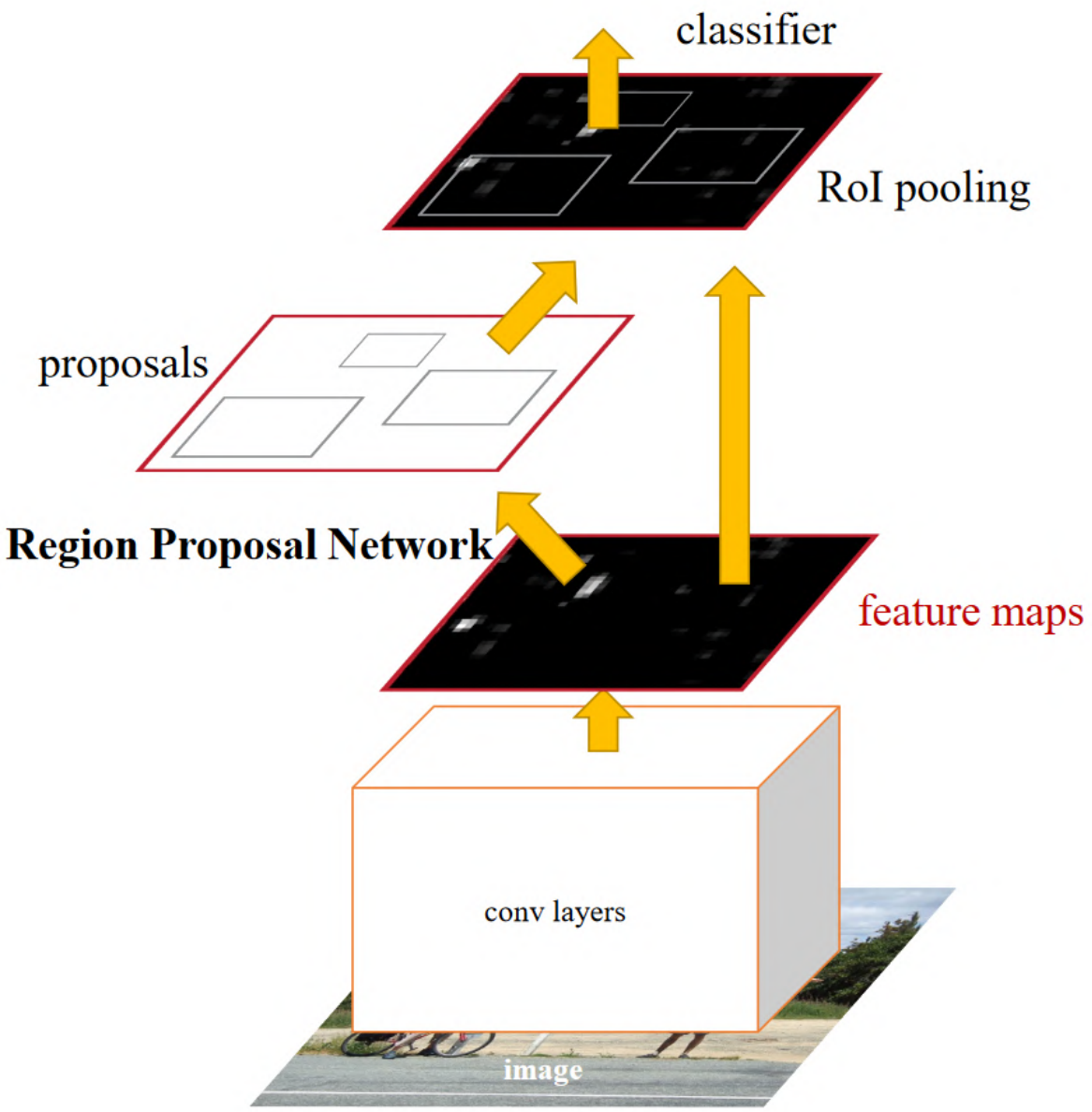


Figure 2.2: Architecture of Faster R-CNN. A convolutional backbone produces feature maps, an RPN generates candidate regions, and a second-stage head classifies and refines these proposals. The figure is taken from the original paper [6]

YOLOv5

YOLOv5 [3] belongs to the "You Only Look Once" family of detectors [7]–[9], [23], which reformulated object detection as a direct regression task. Instead of relying on region proposals or multi-stage processing, YOLO-based models predict bounding boxes and class probabilities in a single pass over the image. This one-stage design makes them significantly faster than two-stage approaches, while still maintaining competitive accuracy.

YOLOv5 builds upon earlier generations by adopting a modern architecture implemented in PyTorch. Its design combines a **CSPDarknet backbone** for feature extraction, a **PANet neck** [21] for multi-scale feature fusion, and lightweight detection **heads** that output predictions at different resolutions. This modular structure enables the detector to capture both coarse semantic information and fine-grained details, which is

especially important for detecting small or subtle abnormalities in medical images. Compared with its predecessors, YOLOv5 also integrates practical training improvements, such as automatic anchor box learning, dynamic image resizing, and strong augmentation strategies including *Mosaic* and *MixUp*. These augmentation methods are explained in detail in Section 2.4.

As shown in Fig. 2.1, PANet enhances the original FPN design by introducing a bottom-up path augmentation that allows information from lower-level features to propagate upward more effectively, improving localization and segmentation performance.

The main advantage of YOLOv5 lies in its efficiency. It achieves high throughput and can process images at real-time speeds while supporting large input sizes. This makes it suitable for scenarios that require rapid decision-making, such as clinical workflows where a large number of chest X-rays must be screened in a short time. At the same time, YOLOv5 inherits some limitations of the YOLO family, such as reduced localization accuracy for very small or overlapping objects compared to two-stage models like Faster R-CNN. Nevertheless, its balance of speed and accuracy has made it one of the most widely adopted detectors in both research and applied contexts.

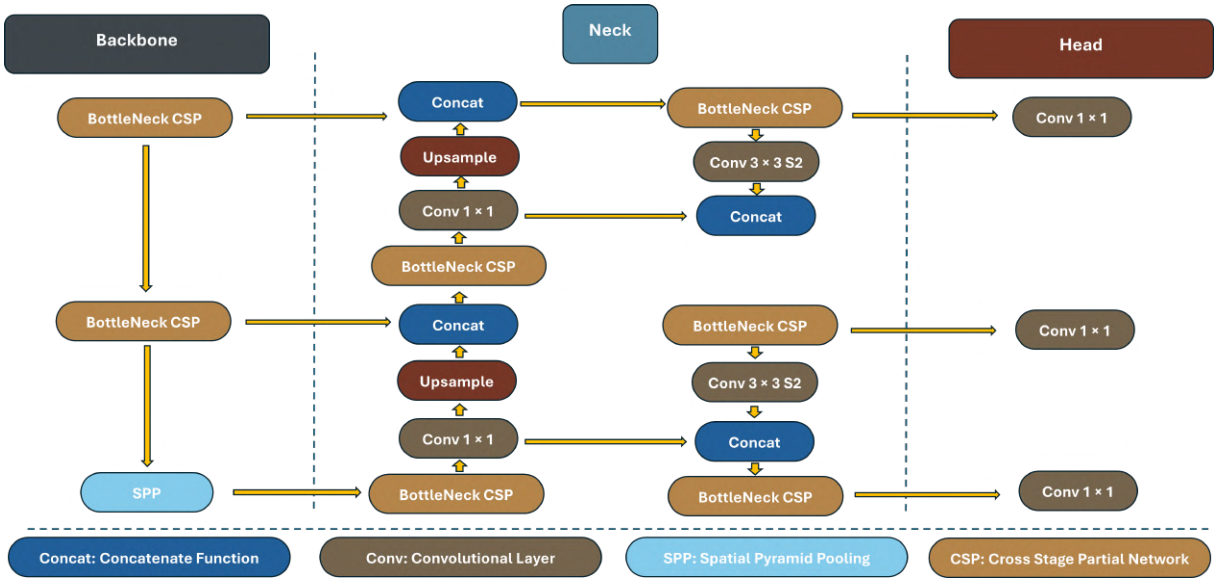


Figure 2.3: Architecture of YOLOv5: **CSPDarknet** backbone, **PANet** neck, and multi-scale detection heads, adapted from [24].

EfficientDet

EfficientDet [4] is a one-stage detector that aims to improve the balance between accuracy and efficiency. It combines the EfficientNet backbone [19] with a BiFPN, designed for effective multi-scale feature fusion. Unlike earlier FPN-based methods, BiFPN introduces weighted connections that allow the network to learn the relative importance of different resolution features. This bidirectional fusion ensures that both low-level details and high-level semantic information are consistently aggregated across scales, which is particularly useful for detecting small and subtle objects such as nodules or calcifications in medical images.

A further innovation is *compound scaling*, which scales the backbone depth, feature network width, and input resolution in a coordinated way. Previous approaches often scaled only one of these dimensions, leading to inefficient trade-offs between performance

and resource demands. EfficientDet's compound scaling strategy enables the creation of a family of models (D0–D7), each tuned to different computational budgets, while maintaining a favorable accuracy–efficiency balance. Additionally, the use of depthwise separable convolutions reduces the overall number of parameters and FLOPs without sacrificing detection quality.

In the VinBigData challenge, EfficientDet (particularly the D2 variant) was widely adopted because it achieved strong accuracy while keeping inference costs moderate. This made it well suited for large-scale clinical datasets, where computational efficiency is critical. EfficientDet illustrates how careful architectural design—through BiFPN for feature fusion and compound scaling for adaptability—can deliver high-quality detection results across a wide range of resource constraints(Fig. 2.4).

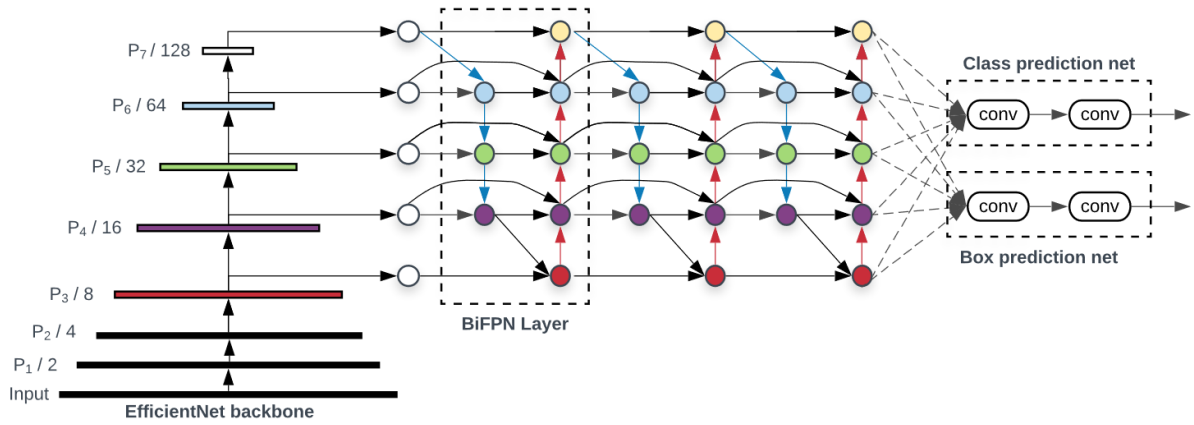


Figure 2.4: EfficientDet architecture combining **EfficientNet** backbone with **BiFPN** for multi-scale feature fusion. Figure taken from [4].

VFNet (VarifocalNet)

VFNet (VarifocalNet) [5] was introduced to address a recurring limitation of dense one-stage detectors: the mismatch between classification confidence and localization accuracy. In conventional detectors, the classification score is used to rank candidate detections, but this score does not necessarily reflect how well a predicted bounding box overlaps with the ground truth. As a result, high-quality boxes may be suppressed during non-maximum suppression, reducing overall detection performance.

To overcome this problem, VFNet predicts an **IoU-aware classification score** that combines both object presence confidence and localization quality. This is achieved using a novel *Varifocal Loss*, which encourages the model to assign higher scores to bounding boxes that are not only classified correctly but also localized accurately. In addition, VFNet integrates the **Adaptive Training Sample Selection (ATSS)** mechanism [25], which improves the stability of positive/negative sample assignment across feature levels. Together, these components make the detector more robust and effective in ranking detections.

VFNet also introduces a **star-shaped bounding box representation**, which leverages deformable convolution to capture the geometry and local context of each box more effectively 2.5. This richer representation helps refine bounding box predictions beyond standard regression heads. Overall, VFNet exemplifies how integrating IoU-aware scoring and adaptive training into a dense, anchor-free framework can yield both higher precision and more reliable detection results. These characteristics are especially valuable in

medical imaging, where distinguishing subtle abnormalities requires consistent ranking of high-quality predictions.

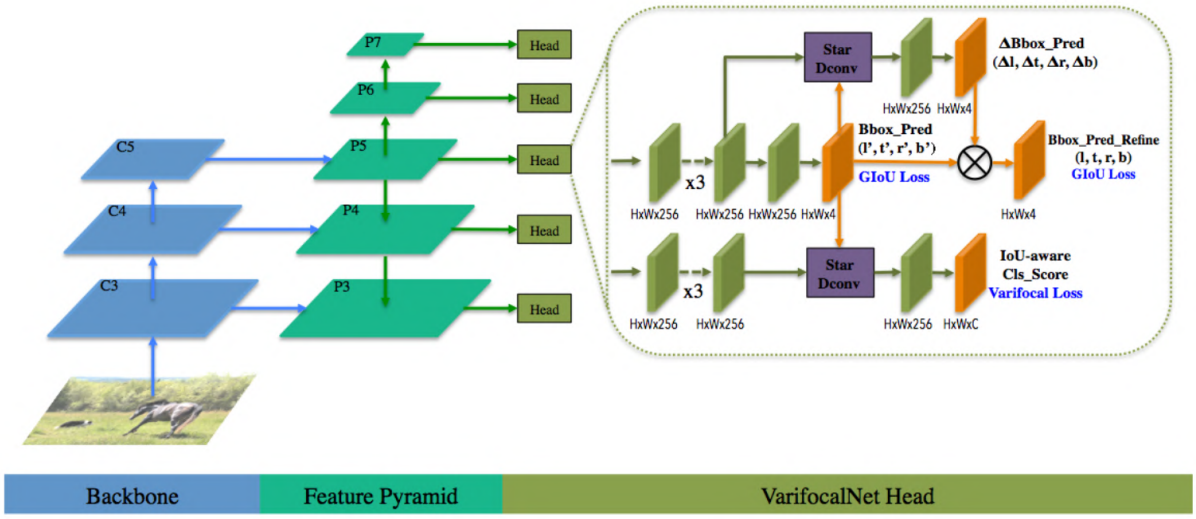


Figure 2.5: VFNet: IoU-aware one-stage detection with Varifocal Loss, ATSS sample assignment, and star-shaped bounding box representation for improved ranking and localization. The figure is taken from the original paper [5]

Summary

In the VinBigData competition, these core architectures were often combined with additional strategies to enhance robustness and improve overall accuracy. Advanced augmentation methods 2.4 played an important role in improving generalization, while post-processing techniques such as WBF were used to merge predictions from multiple models and boost mAP. Some participants also experimented with transformer-based detectors such as DETR [10] and Deformable DETR [26], although these approaches were less common due to higher computational demands and longer training times.

By leveraging complementary strengths—such as the speed of YOLOv5, the efficient scaling of EfficientDet, the IoU-aware predictions of VFNet, and the precise localization of Faster R-CNN—participants were able to address the diverse challenges of chest X-ray abnormality detection and achieve competitive results across a wide range of conditions.

2.3 Transformers for Detection

Transformer-based object detectors introduced a fundamental shift in object detection by reformulating it as a *set prediction* problem. Rather than scanning images with dense anchors or proposals, these models predict a fixed number of objects directly and match them to ground-truth boxes using bipartite (Hungarian) assignment. The first framework to apply this idea, **DETR (DEtection TRansformer)** [10], consists of a convolutional backbone followed by a transformer encoder–decoder that processes a sequence of learned *object queries*. Each query attends to global image features and outputs a class label and bounding box. This design (Fig. 2.6) eliminates hand-crafted anchors and post-processing steps such as non-maximum suppression, offering a clean, end-to-end pipeline. However, the original DETR suffers from slow convergence and reduced performance on

small objects, since global attention makes optimization difficult and ignores multi-scale feature cues.

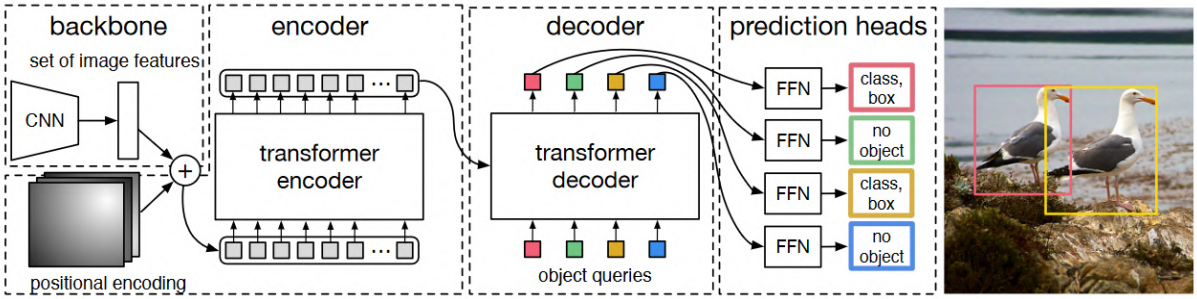


Figure 2.6: Overview of the DETR architecture based on [10]. A CNN backbone extracts features that are processed by a transformer encoder–decoder operating on object queries. Predictions are matched with ground truth using bipartite matching, removing the need for anchors and NMS.

To overcome these limitations, **Deformable DETR** [26] introduces multi-scale deformable attention, where each query attends only to a small, learned set of sampling points around a reference position across several feature levels. This selective attention mechanism greatly reduces computation and accelerates training, while improving small-object detection through multi-scale representation. Deformable DETR retains DETR's end-to-end structure but achieves faster convergence and higher accuracy, becoming the foundation for most later transformer detectors.

Among the recent generation of improved detectors, **Co-DETR (Collaborative DET- R)** and its derivative **Co-DINO** integrate the strongest ideas from the DETR family into a unified framework. These models adopt a collaborative query design that merges ideas from dynamic anchor box queries [27], conditional query decomposition [28], and denoising training [29]. The denoising component introduces slightly perturbed ground-truth boxes during training, allowing the model to learn robust matching and significantly speed up convergence. Co-DETR also employs multi-scale deformable attention for efficient feature fusion and pairs the detection head with hierarchical transformer backbones such as the Swin Transformer [30]. In this thesis, the *Co-DINO variant with a Swin-Large backbone* is used, as it provides stronger spatial reasoning, faster convergence, and higher overall accuracy than the standard DETR-R50 baseline while preserving the end-to-end formulation.

Other DETR variants further refine query design and training stability. *Conditional DETR* [28], *DAB-DETR* [27], and *DN-DETR* [29] focus on query parameterization and denoising; *Anchor DETR* [31] reintroduces anchor-point initialization to guide spatial learning; and efficiency-oriented versions such as *Sparse DETR* [32] and *UP-DETR* [33] address computational and data-efficiency challenges.

Compared with CNN-based detectors described in Sections 2.2 and 2.1, transformer-based models offer a fully differentiable pipeline and the ability to capture long-range dependencies through attention. Their main trade-offs involve heavier computation, dependence on large-scale data, and the need for advanced training strategies. The models applied in this thesis—**DETR-R50** and **Co-DINO (Swin-L)**—represent two stages of this evolution: the first establishes the transformer paradigm for object detection, and the second refines it into a more efficient and accurate framework suitable for large-scale medical imaging tasks.

Alongside these head architectures, transformer backbones have also started to replace CNNs in some pipelines. **ViTDet** [34] is a good example: it keeps the plain Vision Transformer structure without introducing a hierarchical pyramid inside the backbone. Instead, it extracts multi-scale features from later ViT blocks using a lightweight FPN and uses windowed attention to keep computation manageable at high resolutions. ViTDet demonstrates that even a straightforward ViT backbone can compete in object detection, provided it is trained effectively and paired with the appropriate augmentations.

2.4 Augmentation Techniques

Data augmentation plays a crucial role in improving the robustness and generalization of object detection models. Traditional methods, such as random horizontal flips, small rotations, crops, brightness and contrast adjustments, or Gaussian noise addition, remain effective due to their simplicity and low computational cost. However, modern deep learning models—particularly large convolutional and transformer-based detectors—benefit greatly from more advanced augmentation strategies designed to increase the diversity of training samples while maintaining semantic consistency.

MixUp [35] was one of the first techniques to move beyond geometric transformations. It linearly combines two images and their corresponding labels, generating a new synthetic sample that lies between the two in both pixel and label space. By encouraging the model to behave linearly between training examples, MixUp regularizes the decision boundaries and reduces overfitting. Despite its simplicity, MixUp often produces visually ambiguous samples, which can make precise localization more difficult in dense detection tasks.

CutMix [36] addresses this limitation by introducing a spatially grounded alternative. Instead of blending entire images, CutMix replaces a randomly selected rectangular region of one image with a patch from another. The new label is adjusted according to the ratio of pixels taken from each source. This approach retains the regularization benefits of MixUp but produces more realistic composite images, improving both classification and localization. Yun *et al.* (2019) demonstrated that CutMix outperforms MixUp and Cutout on multiple benchmarks, including ImageNet classification and Pascal VOC object detection, and enhances model robustness to occlusion and out-of-distribution inputs.

AutoAugment [37] takes a different perspective by automating the search for optimal augmentation policies. Instead of manually designing transformations, AutoAugment uses reinforcement learning to discover combinations of operations (such as rotation, contrast adjustment, or shearing) that best improve model accuracy on a validation set. This method adapts the augmentation pipeline to the dataset characteristics, though at the cost of increased computational effort during the search phase.

For medical imaging—and chest X-ray datasets in particular—augmentation strategies must also consider domain-specific factors. Synthetic variations such as simulated noise, local contrast enhancement (e.g., CLAHE), or artificial opacities can help mimic differences between scanners, exposure settings, and preprocessing pipelines. Such augmentations improve model robustness to variations across hospitals and imaging conditions, a key requirement for real-world deployment.

In the **VinBigData Chest X-ray Abnormalities Detection Challenge**, augmentations served two main purposes. First, they mitigated class imbalance by generating more diverse examples of rare pathologies. Second, they improved generalization to unseen data by simulating the variability of X-ray acquisition. In this thesis, all reproduced models—including **DETR-R50**, **Co-DINO (Swin-L)**, and CNN-based baselines such

as **YOLOv5**, **EfficientDet**, **VFNet**, and **Faster R-CNN**—were trained using a unified augmentation setup that combined geometric and advanced techniques (MixUp, CutMix, and AutoAugment) to achieve consistent and reproducible results.

2.5 Reproducibility in Machine Learning Research

Reproducibility has emerged as a critical concern in modern machine learning (ML) research, particularly in high-stakes domains such as medical imaging. Inconsistent experimental reporting, missing implementation details, and unavailability of source code can make it difficult—or even impossible—for other researchers to replicate published results. This problem has been widely recognized across the community, with several studies highlighting the "reproducibility crisis" in artificial intelligence (AI) [15], [38], [39].

In the context of object detection for medical imaging, reproducibility challenges are amplified by the complexity of end-to-end pipelines. These often involve numerous hyperparameters, data preprocessing steps, and hardware-specific optimizations that may not be fully documented. As Pineau, Vincent-Lamarre, Fortin, *et al.* [15] and Pineau, Vincent-Lamarre, Sinha, *et al.* [40] note, even small deviations in training procedure or evaluation metrics can lead to substantial differences in reported performance, making fair comparison between methods difficult.

Several taxonomies have been proposed to formalize the concept of reproducibility. For example, Tatman, Vanderplas, and Dane [41] distinguish between *exact reproducibility* (replicating results using the same code and data), *statistical reproducibility* (obtaining equivalent results within expected variability), and *conceptual reproducibility* (validating the main claims using different data or methods). This framework is particularly relevant for this thesis, which seeks to ensure both exact and statistical reproducibility of baseline results, while also testing conceptual reproducibility through alternative architectures and training strategies.

Table 2.1: Taxonomy of reproducibility types in machine learning research, adapted from Tatman, Vanderplas, and Dane [41].

Type	**Definition**
Exact reproducibility	Replicating the same results using the original code, dataset, and computational environment.
Statistical reproducibility	Achieving results within expected statistical variation using the same experimental setup (code, data, protocol).
Conceptual reproducibility	Supporting the same scientific claims using different data, methods, or experimental conditions.

Quantifying the reliability of ML experiments has also been an active area of study. Henderson, Islam, Bachman, *et al.* [42] shows that uncontrolled sources of randomness—such as weight initialization, environment seeds, and non-deterministic hardware operations—can significantly affect performance, especially in deep reinforcement learning. Similarly, Bouthillier, Delaunay, Bronzi, *et al.* [43] emphasizes the need to account for variance when benchmarking ML models, warning that single-number performance reporting can be misleading without statistical context.

Efforts to improve reproducibility have included community-driven initiatives, such as the NeurIPS Reproducibility Program [15], [40], which encourage authors to share code, datasets, and detailed experimental protocols. These initiatives have been shown to improve transparency and facilitate more rigorous comparative studies [44]. However, reproducibility is not solely a matter of openness—model generalization must also be validated across diverse datasets and conditions to ensure robustness, particularly in safety-critical applications like computer-aided diagnosis.

In this thesis, reproducibility is a guiding principle. All reproduced models are implemented in open-source frameworks (MMDetection, MMYOLO, EfficientDet-PyTorch) with standardized, modular pipelines. Detailed configuration files, dataset preparation scripts, and environment specifications are maintained to enable exact re-execution of experiments. Furthermore, each reproduced baseline is validated over multiple training runs to account for random variation in results, following the recommendations of Bouthillier, Delaunay, Bronzi, *et al.* [43]. By adhering to these practices, this work aims to contribute both high-performance detection models and a reliable, reusable experimental framework for future research in medical AI.

Chapter 3

Dataset and Benchmarking Framework

This chapter provides an overview of the data and experimental infrastructure employed in the study. The VinBigData Chest X-ray dataset is introduced first, including details on file formats, class distribution, and the implications of using DICOM versus PNG/JPEG for model training. Subsequently, the preprocessing and annotation consolidation procedures are described, with particular attention given to the application of Weighted Boxes Fusion and stratified cross-validation for producing stable and traceable labels. The benchmarking setup, encompassing the frameworks, computing environment, and training pipeline, is then presented to ensure consistent comparison across models. Finally, the evaluation metrics utilized in this work are outlined, and their correspondence to Kaggle and COCO reporting standards is clarified. Collectively, these components establish a transparent and reproducible foundation for the experiments discussed in subsequent chapters.

3.1 VinBigData Chest X-ray Dataset Description

The dataset used in this thesis is the **VinBigData Chest X-ray Abnormalities Detection dataset**, released as part of the Kaggle competition in 2021. It was collected in several Vietnamese hospitals and carefully annotated by a panel of 17 board-certified radiologists. In total, the dataset consists of **18,000 frontal chest radiographs** labeled with bounding boxes corresponding to 14 abnormality classes, as well as a "No finding" category. The official competition split provided 15,000 images for training, 1,500 images for validation, and 1,500 for testing, which is the split adopted in this thesis [45].

DICOM format

A distinctive aspect of this dataset is the way the images were released. The original radiographs were stored in the **DICOM (Digital Imaging and Communications in Medicine)** format, which is the global standard for handling and exchanging medical imaging data. DICOM files are not limited to the image itself but include a wide range of metadata: information about the patient (such as age and sex), acquisition parameters (such as X-ray tube voltage, exposure settings, and detector type), as well as hospital and device identifiers. This combination of pixel data and structured metadata allows DICOM to serve as a self-contained clinical object, ensuring interoperability across institutions and vendors [46], [47]. Because of these properties, DICOM plays a critical role in radiology

workflows, enabling consistent storage, communication, and retrieval of medical images in Picture Archiving and Communication Systems (PACS).

At the same time, DICOM poses certain challenges when used for machine learning. Unlike consumer formats such as JPEG or PNG, DICOM images may be encoded with varying bit depths (commonly 12- or 16-bit grayscale, rather than the 8-bit depth of standard consumer images). This higher precision preserves subtle intensity variations that are important for diagnostic purposes but requires careful normalization before training neural networks. Furthermore, DICOM images are often very large, and the embedded metadata can make them difficult to process efficiently at scale. Differences in how vendors implement optional parts of the DICOM standard can also result in inconsistencies, as highlighted in prior studies [46].

Comparison with JPEG and PNG

For broader accessibility, Kaggle also provided **JPEG** and **PNG** versions of the dataset. These formats are easier to handle in typical machine learning pipelines but come with important trade-offs. JPEG, formalized in the international standard ISO/IEC 10918-1:1992, is based on lossy compression [48]. By discarding high-frequency details, JPEG achieves much smaller file sizes, but it also introduces compression artifacts. In the context of chest radiographs, such artifacts may obscure fine structures or subtle lesions, which are precisely the features radiologists and detection algorithms need to identify. Even mild compression can therefore pose a risk for diagnostic fidelity.

PNG, in contrast, is a **lossless compression format** standardized as ISO/IEC 15948: 2003 [49]. It preserves the full pixel values of the original image, ensuring that no diagnostic information is lost during conversion. PNG also supports higher bit depths than JPEG, which makes it more appropriate for medical imaging. However, PNG files are typically larger than their JPEG counterparts, which increases storage requirements and slows down data transfer and training pipelines. In practice, many medical imaging studies that rely on public datasets prefer PNG over JPEG, because it offers a balance between fidelity and usability, though it still lacks the metadata richness that makes DICOM valuable in clinical settings.

In summary, DICOM provides a complete and metadata-rich representation of clinical images but is complex to handle computationally. PNG preserves image quality faithfully but strips away the metadata, while JPEG reduces storage requirements at the cost of introducing irreversible artifacts. For the VinBigData dataset, this trade-off means that researchers must choose between preserving full clinical fidelity (using DICOM) and simplifying processing (using PNG or JPEG). In this thesis, the converted PNG format is primarily used for experiments, but the implications of this choice are critically acknowledged.

Class distribution

Another important characteristic of the dataset is the **imbalance in class distribution**. Some categories are represented by only a handful of positive cases, such as "Clavicle fracture" with a single labeled image, while others, such as "Aortic enlargement" and "Other diseases," have thousands of instances. The largest group in the dataset is the "No finding" class, which includes over 10,000 images without any abnormalities. Figure 3.1 shows the distribution of positive images across the abnormality classes, reproduced from Nguyen et al. [45]. This imbalance is a central challenge in training robust object detection

models, since models may overfit to frequent classes while failing to generalize to rare conditions.

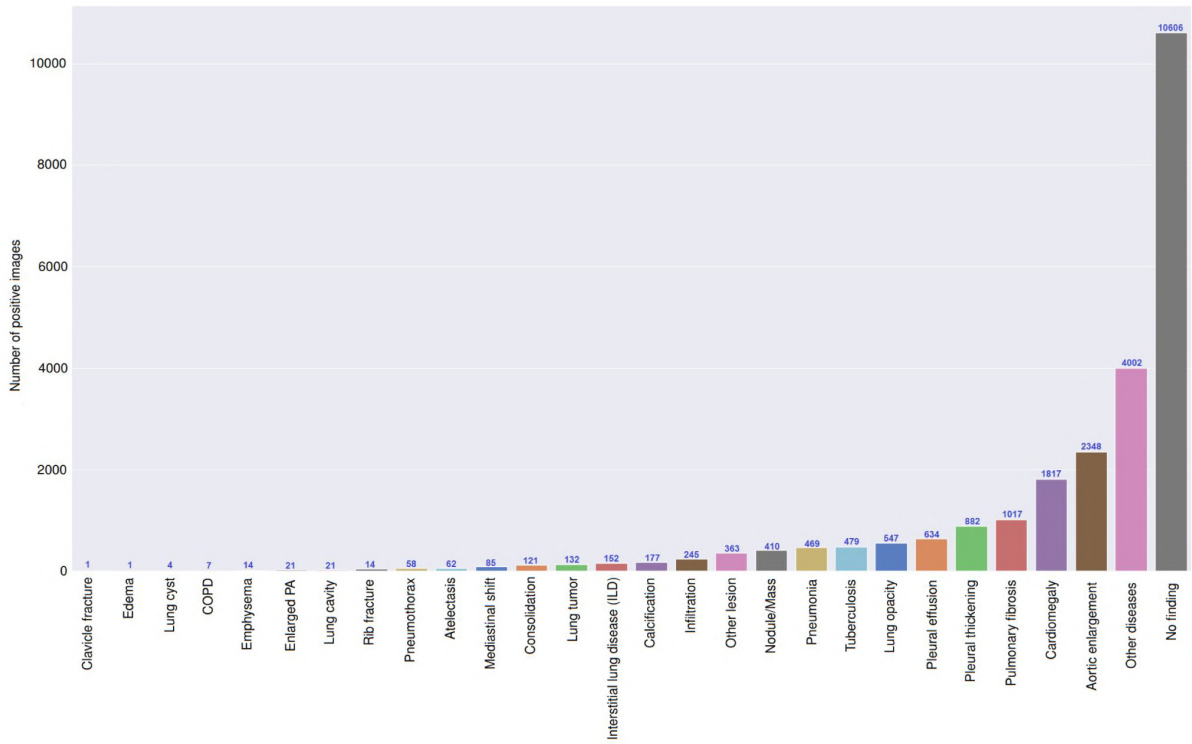


Figure 3.1: Distribution of positive images across abnormality classes in the VinBigData Chest X-ray dataset, adapted from [45].

In addition to per-class frequencies, it is informative to examine the *number of unique abnormality classes per image* in the training set (Fig. 3.2). Most images contain *no bounding boxes* ("No finding"), and the majority of positive images contain *one* or *two* distinct classes, with a long tail of cases showing multiple concurrent findings. This pattern confirms that multi-label scenarios exist, which affects learning dynamics (e.g., class co-occurrence, label imbalance) and motivates the use of augmentations and quality-aware objectives in later chapters. The validation split exhibits a similar distribution and is therefore omitted here for brevity.

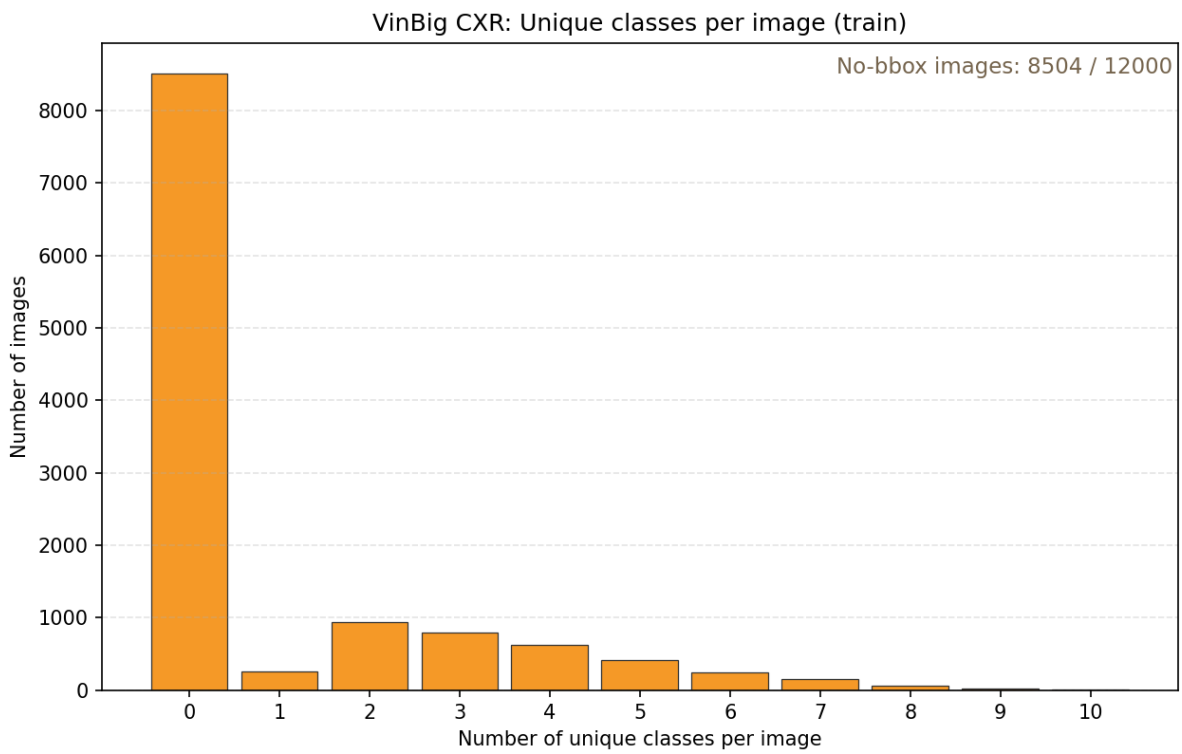


Figure 3.2: Histogram of the number of unique abnormality classes per image in the **training** split of VinBigData CXR. The dominant bar at zero corresponds to "No finding" images; most remaining images contain one–two classes, with a decreasing tail for higher counts.

Image preprocessing

To prepare the images for training, a standardized pipeline was established across frameworks, with adjustments tailored to model requirements. YOLOv5 used letterboxed resizing to 640×640 pixels, combined with multi-scale training at smaller resolutions (320×320 and 480×480), reflecting the default strategy of MMYOLO [3]. EfficientDet-D2 applied random scaling and padding to a fixed size of 768×768, ensuring a balance between accuracy and efficiency [4]. Faster R-CNN and VFNet operated at 512×512 pixels, while transformer-based detectors such as DETR-R50 and Co-DETR Swin-L were trained at 800×800 pixels to exploit their global attention mechanisms [10]. Since chest radiographs are inherently grayscale, each image was replicated into three channels to remain compatible with ImageNet-pretrained backbones. Normalization was performed using the ImageNet mean and standard deviation ([123.675, 116.28, 103.53] and [58.395, 57.12, 57.375], respectively), a step that aligns input distributions with the assumptions of pretrained models and improves convergence.

Annotation preprocessing

A key challenge in the VinBigData dataset lies in the **variability among radiologist annotations**. Even for the same image, bounding boxes for a given abnormality may differ slightly in position, size, or labeling. This inter-observer variation is a well-documented issue in medical imaging [1], [50] and, if unaddressed, can negatively affect both reproducibility and model reliability.

To mitigate this issue, the **Weighted Boxes Fusion (WBF)** method [14] was applied with an IoU threshold of 0.4. WBF merges overlapping bounding boxes from multiple annotators into a single consensus box, weighting their contributions by confidence. During implementation, bounding boxes were normalized to [0,1] coordinates for fusion and then restored to pixel-based COCO xywh format. Each annotation retained its radiologist identifiers (`rad_ids`) and included rectangular segmentations aligned with the bounding boxes, preserving traceability.

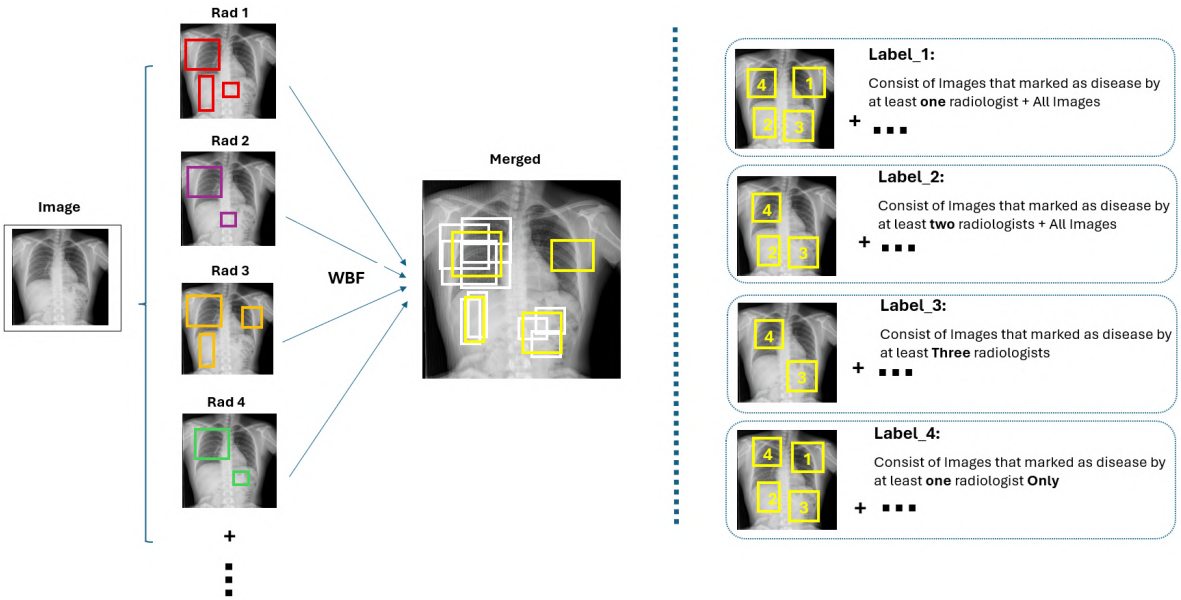


Figure 3.3: Illustration of the WBF process and annotator agreement–based datasets. Left: bounding boxes from multiple radiologists are merged into a single consensus label using WBF. Right: creation of four dataset variants based on annotation agreement thresholds (e.g., images labeled by at least one, two, or three radiologists). These variants were used to reproduce the fourth-place competition solution.

Based on WBF and annotator consensus, three dataset configurations were created:

1. **WBF + Cross Validation (CV) folds:** After WBF, the dataset was split into multiple folds for cross-validation.

2. **WBF + train/validation split:** After WBF, an 80/20 split was applied based on filenames to maintain reproducibility.

3. **Annotator agreement + CV folds:** Four datasets were constructed according to agreement thresholds ($\geq$1, $\geq$2, all three, or $\geq$1 restricted to diseased images). This configuration was specifically used for reproducing the **4th-place competition solution**.

To ensure balanced model evaluation, a **stratified cross-validation** scheme was adopted, maintaining similar class distributions across folds. This prevents rare conditions from being overrepresented or absent in certain validation sets. Figure 3.4 and 3.5 provide a schematic overview of the training and validation process for 4 CV folds.

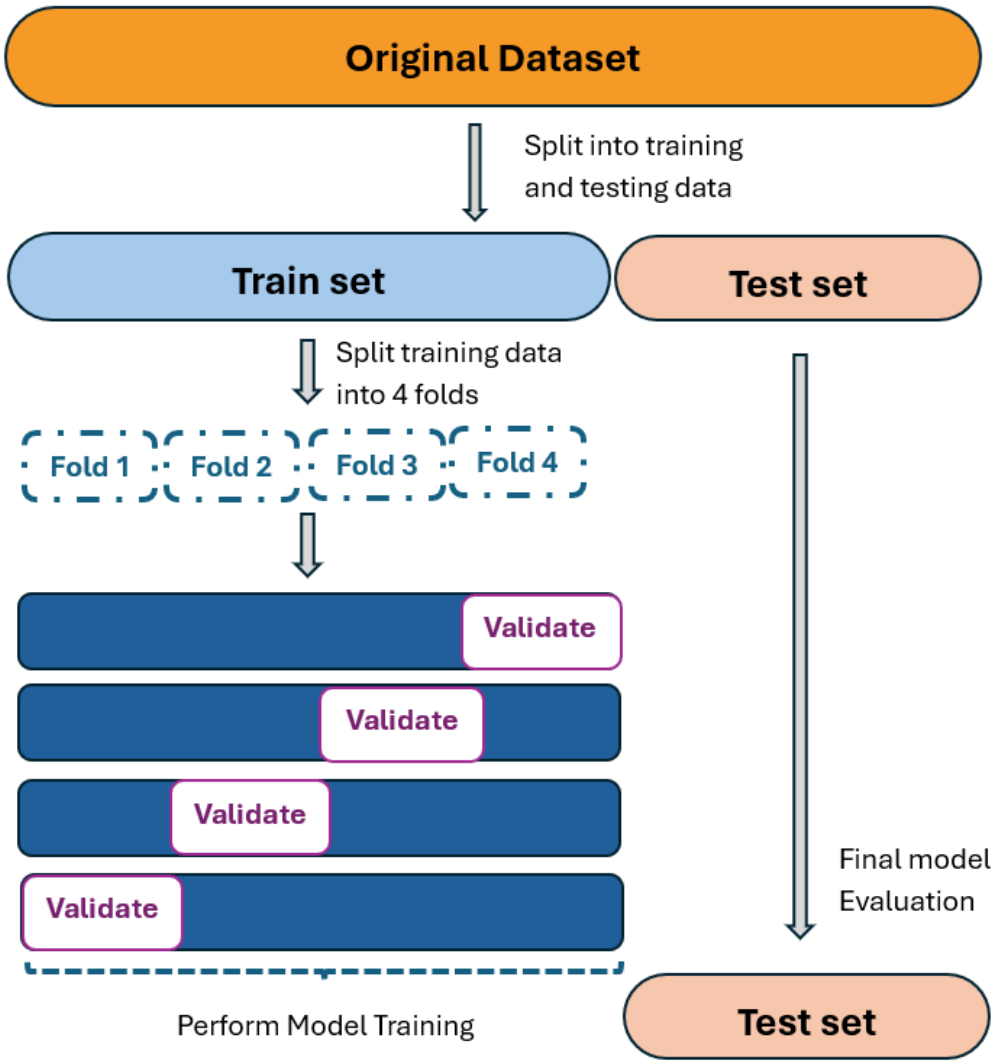


Figure 3.4: Cross-validation workflow for 4 CV folds. The dataset is divided into training and testing sets, followed by a four-fold split of the training data. Each fold is used once for validation, while the remaining folds are used for training, ensuring robust performance estimation.

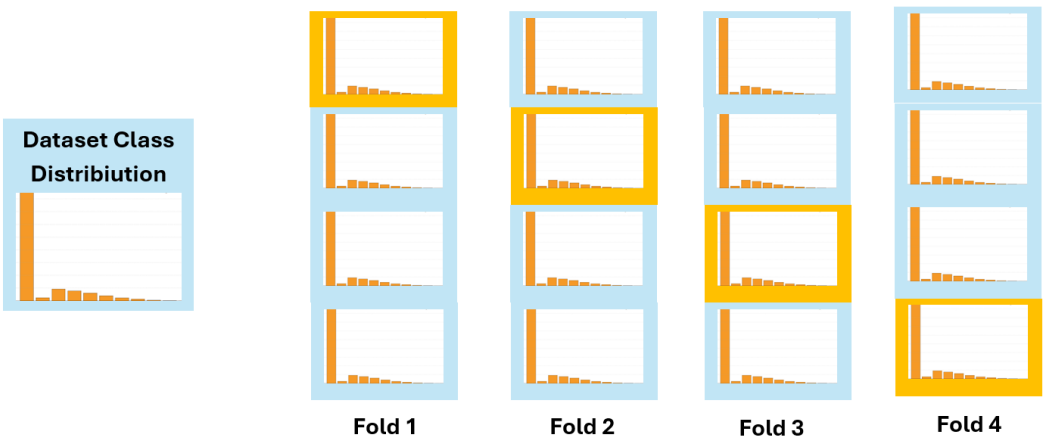


Figure 3.5: Example of stratified cross-validation. Each fold maintains a class distribution similar to the overall dataset, ensuring that rare abnormality classes are represented in all folds. This approach enhances stability and fairness in model evaluation.

Even after applying WBF, minor differences in bounding box placement persisted. Figure 3.6 illustrates representative cases where different radiologists annotated the same abnormality with slightly varying boxes. While these discrepancies may seem small, they highlight the inherent subjectivity of medical image interpretation, motivating the use of WBF to produce consistent consensus labels. More severe disagreements, such as differing opinions on the *type* of abnormality (e.g., interstitial lung disease vs. pulmonary fibrosis), are discussed in Chapter 6.

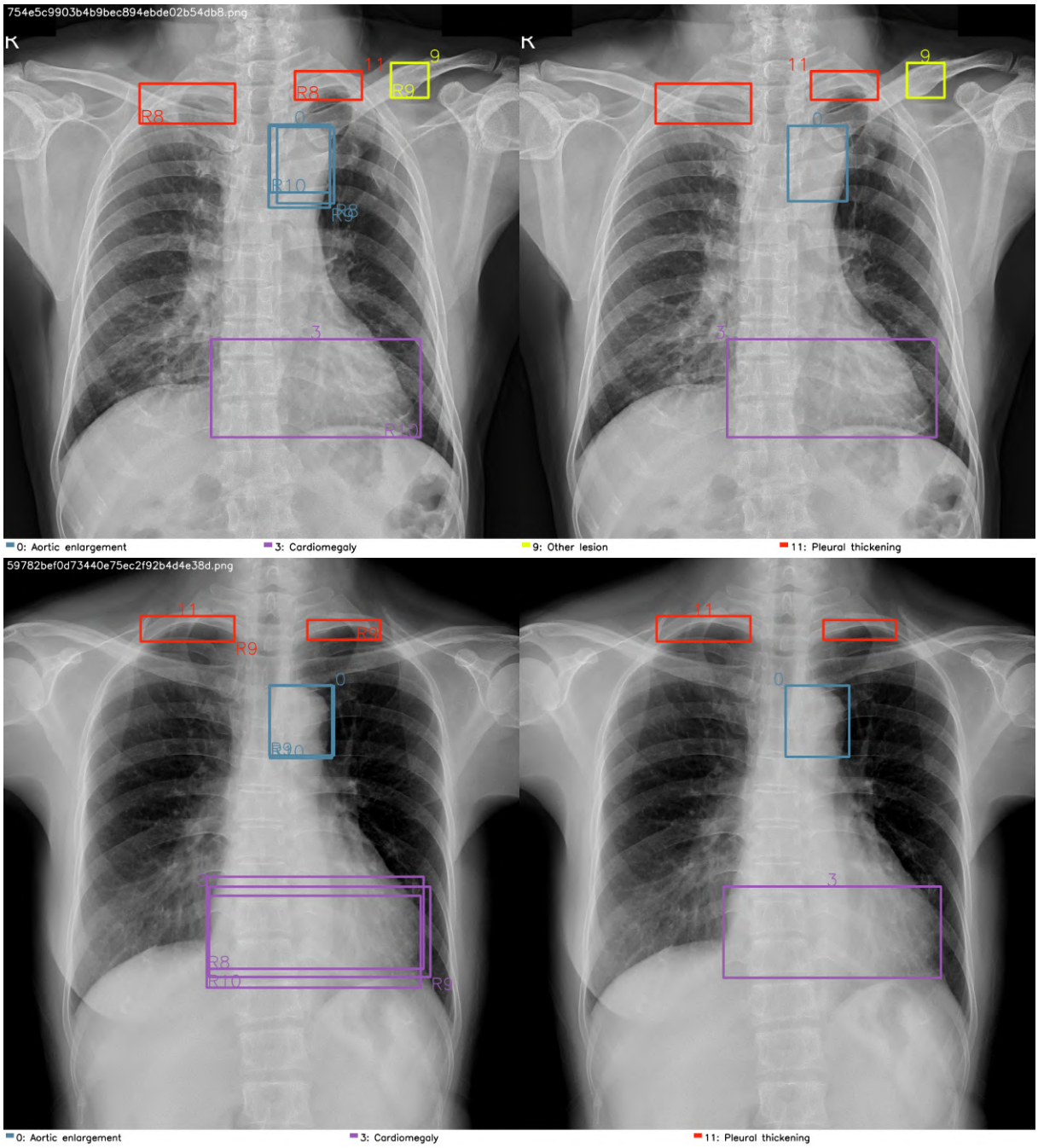


Figure 3.6: Examples of annotation differences between radiologists before and after WBF. Each row shows one case, with annotations from different radiologists on the left and the fused consensus result on the right.

Augmentation strategies

Data augmentation was essential to improve generalization and address the dataset's extreme class imbalance. Simple augmentations included random horizontal flipping (p=0.5), aspect ratio–preserving resizing, and photometric distortions (brightness, contrast, saturation, and hue). Such techniques are well established in medical imaging to increase robustness [51].

More advanced augmentations varied across frameworks. In YOLOv5, Mosaic aug-

mentation combined four images into one grid, enhancing diversity and context, while MixUp blended pairs of images and labels to regularize training [35]. CLAHE (contrast-limited adaptive histogram equalization) was also occasionally applied to improve local contrast, which can be especially useful in X-rays where lesions may appear subtle. EfficientDet employed RandomResizePad and horizontal flips, leveraging compound scaling for balanced accuracy and efficiency [4]. Transformer-based detectors (DETR-R50 and Co-DETR Swin-L) combined resizing with horizontal flips and photometric distortions, but did not use heavier augmentations such as CopyPaste or Large-Scale Jittering due to time constraints. Faster R-CNN and VFNet used resize, flip, and moderate color augmentations.

Finally, test-time augmentation (TTA) was investigated, particularly for YOLOv5. While it occasionally improved validation results through multi-scale inference and flipping, the gains were inconsistent for other models, and TTA was therefore only used for Yolov5 models to improve the performance.

In summary, preprocessing in this thesis was designed to strike a balance between reproducibility and robustness. WBF provided more consistent annotations by reducing inter-radiologist noise, while carefully selected augmentations enhanced model generalization without introducing unrealistic distortions. Together, these steps ensured that subsequent experiments in Chapter **??** and beyond were built on a stable and transparent foundation.

3.2 Benchmarking Frameworks

The experimental setup for this thesis relied on three open-source object detection frameworks: **MMDetection** [16], **MMYOLO** [17], and **EfficientDet-PyTorch** [52]. The use of multiple frameworks was intentional. While MMDetection and MMYOLO provide a unified platform for most convolutional and transformer-based detectors, EfficientDet-PyTorch was included to support the EfficientDet-D2 model family, which was not natively integrated into the other frameworks. This combination ensured that all models selected for reproduction in Phase 1, as well as new architectures explored in Phase 2, could be trained and evaluated in a consistent environment. Another advantage of using COCO-compatible frameworks was that the VinBigData dataset, after preprocessing, was stored directly in **COCO JSON format**, avoiding the need for custom data loaders or annotation conversions.

Computing environment

All experiments were conducted on a Linux cluster running Ubuntu 22.04.5 LTS. The setup used **CUDA 12.5** (driver/runtime) with the **CUDA toolkit 11.7**, and **PyTorch 1.12.1+cu113** built against CUDA 11.3. Two **NVIDIA Tesla V100-PCIE GPUs** with 16 GB VRAM each were available, providing sufficient memory for training both lightweight convolutional models and large transformer-based architectures. During training, approximately 16 GB of VRAM per GPU was accessible, allowing batch sizes and model choices comparable to those reported in the original competition solutions.

Table 3.1: Computing environment used for all experiments.

Component	**Specification**
Operating System	Ubuntu 22.04.5 LTS
CUDA (driver/runtime)	12.5
CUDA Toolkit (nvcc)	11.7
PyTorch version	1.12.1+cu113 (built with CUDA 11.3)
GPUs	2 × NVIDIA Tesla V100-PCIE
GPU memory	16 GB VRAM per GPU (16 GB accessible)

Training pipeline

Each framework supported a modular pipeline, with built-in hooks for logging, checkpoint saving, and validation. For MMDetection and MMYOLO, the default logging style was used, with **Weights & Biases (WandB)** [53] available as an optional integration for experiment tracking. EfficientDet-PyTorch implemented a similar logging system, modeled after MMDetection.

Checkpoint saving and validation strategies followed the original solutions where possible. For solutions in which validation strategies were explicitly reported, the same procedures were replicated. For solutions without available validation details, a consistent approach was applied: validation was performed at the end of every epoch, and in addition to the best epoch checkpoint, the last five checkpoints were saved to allow for later ensembling or ablation studies. In Phase 2 experiments, validation was scheduled less frequently due to time constraints—typically every two or every five epochs depending on the model size and training duration.

Reproducibility measures

To enhance reproducibility, all experiments with MMdetection models were run with a fixed random seed of `0` and for the EfficientDet-D2 model with a fixed seed of 42. All configuration files and scripts will be made publicly available in the project's repository, ensuring that the results presented in this thesis can be reproduced and extended by future research. A direct repository link will be provided in the final version of this document.

In summary, the benchmarking framework combined flexibility (via multiple frameworks), computational reliability (via a GPU-enabled cluster), and transparency (via open-source configuration files). These choices ensured that both Phase 1 reproductions and Phase 2 extensions were executed under controlled and comparable conditions.

3.3 Evaluation Metrics

A central component of object detection research is the evaluation metric. In this thesis, performance was assessed using **mean Average Precision (mAP)**, a standard measure widely used in the computer vision community.

Intersection over Union (IoU)

The evaluation begins with the calculation of the **Intersection over Union (IoU)** between a predicted bounding box and a ground truth box:

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

A prediction is considered correct if the IoU between predicted and ground truth boxes exceeds a chosen threshold. Figure 3.7 illustrates the concept, where the green region represents the overlap between two bounding boxes.

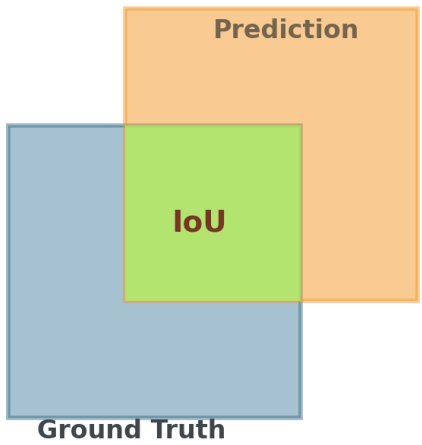


Figure 3.7: Illustration of Intersection over Union (IoU) between two bounding boxes.

Precision and Recall

From the IoU definition, predictions can be categorized into true positives (TP), false positives (FP), and false negatives (FN). This allows defining:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}$$

Precision measures how many predicted detections are correct, while recall measures how many ground truth objects are successfully detected.

Average Precision (AP)

The trade-off between precision and recall is summarized in the **Precision–Recall (PR) curve**. The **Average Precision (AP)** for a given class is defined as the area under this curve:

$$\mathrm{AP} = \int_0^1 p(r)\, dr$$

where $p(r)$ denotes the interpolated precision at recall level r.

Mean Average Precision (mAP)

The final metric, mAP, is obtained by averaging AP across all object classes:

$$\mathrm{mAP} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{AP}_i$$

where N is the number of classes. The choice of IoU threshold(s) strongly affects the reported mAP.

Evaluation in the VinBigData challenge

The **VinBigData Chest X-ray Abnormalities Detection challenge** followed the **PASCAL VOC 2010 protocol** [54], using mAP at a single IoU threshold of IoU > 0.4. Thus, detections were counted as correct if they overlapped with ground truth boxes by more than 40%.

In Kaggle competitions, submissions are evaluated on two separate subsets of the test data: the **public** and **private leaderboards**. The public leaderboard score is computed on a small portion of the hidden test set and is displayed to participants during the competition. It provides an initial indication of model performance but can be affected by overfitting if teams tune their models too closely to this visible subset. The **private leaderboard**, in contrast, is calculated on the remaining, unseen portion of the test set and remains hidden until the competition closes. It serves as the final benchmark for ranking participants and reflects the true generalization ability of each model. A strong discrepancy between public and private scores often indicates overfitting or limited robustness, whereas consistent performance across both leaderboards suggests good model generalization and reliable training practices.

Evaluation in COCO and MMDetection

The COCO dataset introduced a stricter evaluation protocol [55], where mAP is averaged across multiple IoU thresholds from 0.50 to 0.95 in steps of 0.05. In the MMDetection model zoo, the commonly reported **"box AP"** corresponds to this COCO-style mAP@[0.50:0.95], not AP at a single IoU threshold. Specific metrics such as AP50 or AP75 are typically reported separately in training logs.

Metrics used in this thesis

To remain consistent with the challenge, this thesis reports **AP50** (IoU threshold = 0.5) as the primary metric for experimental results. This provides a slightly stricter criterion

than the Kaggle evaluation (IoU = 0.4) while remaining comparable. For the selection of transformer-based models in Phase 2, however, **box AP** values from the MMDetection model zoo were also considered, as they reflect state-of-the-art performance under COCO standards.

Chapter 4

Methodology

The workflow of this thesis was designed to balance reproducibility with exploration. It begins with **dataset preparation**, where the original VinBigData chest X-ray images in DICOM format were converted to PNG, and annotations were stored in COCO-compatible JSON files. A **preprocessing stage** then consolidated the labels provided by multiple radiologists using WBF and, when required, created additional agreement-based label sets. These steps ensured that the final dataset could be used reliably across different model families.

The processed dataset was integrated into **three open-source frameworks**, MMDetection [16], MMYOLO [17], and EfficientDet-PyTorch [52]. Each framework was chosen for its ability to support COCO annotations and for the diversity of models it enabled. This unified setup allowed both established Kaggle solutions and more recent transformer-based architectures to be trained and evaluated under the same experimental conditions.

From this foundation, the methodology proceeded in two phases. **Phase 1** focused on **production of top competition solutions**, specifically the first, fourth and tenth place entries. Wherever training and validation details were available, they were carefully replicated; when details were missing, a consistent validation protocol was applied. These experiments provided the baseline results that anchor the remainder of the thesis.

Phase 2 expanded the scope by **exploring new architectures and training strategies**. Transformer-based detectors such as DETR and Co-DETR Swin-L were implemented, alongside experiments with different augmentation pipelines and validation schedules. These investigations not only demonstrated the adaptability of the benchmarking framework but also offered insights into the performance of newer model families in the medical imaging domain.

Finally, all experiments from both phases were assessed using a consistent set of **evaluation metrics**, namely mean Average Precision (mAP) at various Intersection-over-Union (IoU) thresholds, as detailed in Section 3.3. This ensured that results could be compared fairly across different models and configurations.

4.1 Phase 1: Reproducing Top Kaggle Solutions

In the first phase of this thesis, three high-ranking solutions from the VinBigData Chest X-ray Abnormalities Detection challenge were reproduced. These correspond to the **first**, **fourth**, and **tenth place** entries on the publicly available leaderboard based on their Private Score. The goal was to replicate their reported setups as closely as possible, in order to establish reliable baselines for comparison with later experiments.

1st Place Solution

The winning entry was built on an extensive ensemble of models, reflecting a strategy of maximizing diversity across architectures and training setups. The authors reported training more than a dozen models, which they categorized into three stages of validation:

- **Fully validated stage:** three core models were validated carefully and used as the backbone of the solution:
  1. Faster R-CNN with ResNet-101 backbone (implemented in Detectron2),
  2. YOLOv5,
  3. EfficientDet-D2.
- **Partially validated stage:** a five-fold cross-validation of YOLOv5 models, providing additional robustness but not all folds were equally validated.
- **Not validated stage:** a large number of experimental runs, including two more EfficientDet-D2 models, three YOLOv5 variants (with and without Test-Time Augmentation (TTA)), a YOLOv5 trained on all 16 classes, a five-fold YOLOv5 model shared by another participant, a YOLOv5 with modified anchors, a Detectron2 ResNet-50, and a YOLOv5 with input size 640.

From this large pool, the final submissions were constructed using ensembles of the models that performed best either on local cross-validation (around 0.47+ mAP), on the public leaderboard (0.354), or on the private leaderboard (0.321 best run, 0.314 from the best leaderboard model).

This solution demonstrates how model ensembling at scale can be leveraged to boost leaderboard performance: combining a strong two-stage detector (Faster R-CNN), a lightweight one-stage detector (YOLOv5), and an efficiency-optimized detector (EfficientDet) ensured complementary strengths across models. Where training or validation details were not fully available, this reproduction followed the consistent protocol of validating at each epoch and saving both the last five and the best checkpoints.

Fourth Place Solution

The 4th place solution focused strongly on the **preprocessing of annotations** and systematic use of TTA. The author trained **YOLOv5x** exclusively, but constructed multiple label sets to reflect varying levels of annotator agreement:

- **labels-1:** bounding boxes retained if marked by at least one radiologist, Model-1 is the model trained on this subset of the dataset.

- **labels-2:** bounding boxes retained if marked by two or more radiologists, Model-2 is the model trained on this subset of the dataset.
- **labels-3:** bounding boxes retained only if marked by all three radiologists, Model-3 is the model trained on this subset of the dataset.
- **labels-4:** equivalent to labels-1, but restricted to images containing disease. Model-4 is the model trained on this subset of the dataset.

At the inference stage, the author implemented a modified TTA pipeline. Predictions were generated on six augmented image variants using three different scales $[1.0, 0.83, 0.67]$ in combination with horizontal flipping. The final predictions were merged using **WBF** with IoU=0.5.

The contribution of each label group was carefully evaluated through ensembling experiments:

- model-1: public 0.264, private 0.278
- model-1 + model-2: public 0.271, private 0.295
- model-1 + model-2 + model-3: public 0.267, private 0.303
- model-1 + model-2 + model-4: public 0.278, private 0.297
- model-1 + model-2 + model-3 + model-4: public 0.276, private 0.306

For the final submission, two ensembles were selected:

1. model-1 + model-2 + model-4: public 0.278, private 0.297,
2. model-1 + model-2 + model-4 + outputs from a public notebook: public 0.285, private 0.303.

Ensembles including model-3 were avoided in the final submission because the low sample size of minor diseases in its training set degraded leaderboard performance. This solution demonstrates that careful management of noisy labels and strategic inference augmentations can yield competitive results even with a single detector architecture.

Tenth Place Solution

The Tenth place solution, credited to "kiet-chu," combined multiple detectors into a large ensemble, with a strong emphasis on **model diversity** and **weighted ensembling**. In total, the solution included **twelve trained models**:

- 5 YOLOv5 models, trained on 768×768 images using five-fold CV,
- 1 Faster R-CNN model, implemented in Detectron2 ,
- 6 VFNet models, trained with five-fold cross-validation plus one additional random split.

During preprocessing, **WBF** was applied to unify radiologist annotations. At inference, predictions from all twelve models were ensembled using WBF with IoU=0.5, with model weights scaled according to their AP50 performance. An additional filtering step was introduced by applying a skip-box threshold of 0.3 to reduce noise.

The author also experimented with classification-based approaches but concluded that the solution already available remained stronger than custom-trained classifiers, and thus retained it.

This solution illustrates how a carefully weighted ensemble of heterogeneous detectors—combining YOLOv5, VFNet, and Faster R-CNN—can achieve robust performance by leveraging complementary strengths of each model family. The choice to weight models according to their AP50, rather than treating all equally, highlights the importance of quantitative calibration in ensemble design.

Table 4.1: Summary of Phase 1 reproduced solutions and their leaderboard results on the VinBigData challenge.

Solution	**Models Used**	**Public LB**	**Private LB**
1st place	Faster R-CNN (ResNet-101), YOLOv5, EfficientDet-D2	0.354	0.314
4th place	YOLOv5x	0.285	0.303
10th place	YOLOv5, VFNet, Faster R-CNN	∼0.27	∼0.293

4.2 Phase 2: Exploring New Architectures

While Phase 1 of this thesis was devoted to reproducing top Kaggle solutions, Phase 2 explored the potential of more recent architectures that were not available or widely adopted at the time of the competition. Due to time constraints, only two transformer-based detectors were investigated: **DETR with a ResNet-50 backbone (DETR R50)** and **Co-DETR/Co-DINO with Swin-Large backbone (co_dino_5scale_swin_large_16e)**.

The rationale behind these selections was twofold. First, both models were readily available through the MMDetection framework, allowing them to be integrated into the existing benchmarking pipeline with minimal additional engineering. Second, they represent substantially different points on the spectrum of transformer-based detection:

- **DETR R50** [10] was the original Detection Transformer, introducing end-to-end object detection via bipartite matching and eliminating the need for anchors and non-maximum suppression. On the COCO benchmark, it achieves **39.9 box AP**, which, although lower than the best CNN-based detectors, makes it a useful baseline for transformer-based approaches.

- **Co-DETR/Co-DINO with Swin-Large backbone** [56] represents a more advanced generation of transformer detectors. By incorporating collaborative denoising queries and multi-scale feature interaction, it achieves state-of-the-art performance on COCO, with **64.1 box AP**. This makes it an ideal candidate for evaluating whether the most competitive transformer-based designs can transfer effectively to the medical imaging domain.

In this thesis, both models were trained under the same standardized preprocessing and benchmarking setup described in Chapter 3.2. Their results are reported alongside the Phase 1 baselines, providing insight into whether transformer-based architectures can surpass CNN-based solutions in chest X-ray abnormality detection.

Chapter 5

Proof of Concept and Baseline Results

This chapter presents the baseline experiments that establish the foundation for all subsequent analyses. The primary objective is to reproduce and validate individual object detection models under controlled and transparent conditions before introducing more advanced configurations and ensemble strategies. Each detector is trained and evaluated independently using standardized preprocessing, annotation fusion, and cross-validation procedures derived from the VinBigData Chest X-ray dataset. The results serve as reproducible reference points, enabling fair comparison between architectures and ensuring methodological consistency. The chapter first outlines the experimental setup adopted for baseline reproduction, followed by a detailed presentation of model-specific results and a synthesis of the key insights gained. Together, these experiments confirm the reliability of the benchmarking framework and provide a robust starting point for the extended investigations discussed in the later chapters.

5.1 Experimental Setup for Baseline Reproduction

This chapter establishes reproducible baselines by training and validating each detector *individually*, rather than reconstructing the large competition ensembles. The aim is to fine-tune each model under a clean and measurable protocol before ensembling in later chapters.

The dataset was used in several preprocessed forms: (i) WBF–consolidated annotations with stratified cross-validation (k=5 folds for most runs), (ii) a filename-based train/validation split (default 80/20), and (iii) agreement-filtered label sets (label-1/2/3/4; k=4 folds for the agreement experiments). WBF was applied with internal box normalization to $[0, 1]$ and an IoU threshold of 0.4, then saved back as COCO pixel `xywh`.

All images were used in PNG format and converted to 3-channel RGB where needed (grayscale replicated to three channels). Models followed the framework defaults for input scaling and normalisation unless specified in the configuration files (e.g., EfficientDet-D2 at 768×768 with ImageNet mean/std).

Baseline training used lightweight default configurations to check the convergence of the models. Later on, some basic augmentations which were appropriate for medical data (e.g., random horizontal flip, photometric distortion) were also tested. For YOLOv5, additional experiments included Mosaic/Affine during training and multi-scale TTA at inference.

All runs were evaluated with the COCO metric, and **AP50** (IoU = 0.5) was adopted as the principal comparison criterion, providing a stricter but comparable measure relative to the challenge's mAP@0.4 (Section 3.3).

Unless a solution's original validation routine was documented and followed, models were validated at each epoch, with both the best checkpoint and the last five epochs retained. For YOLOv5 models, which require a higher number of epochs for convergence, models were validated every 5th epoch.

5.2 Results of Baseline Models

As mentioned earlier, baseline experiments in this thesis were designed to establish reproducible reference points before attempting more advanced strategies. Unlike the Kaggle competition submissions, which relied heavily on large ensembles, the focus here was on training and validating each architecture individually. This approach made it possible to evaluate the relative strengths and weaknesses of each detector under consistent experimental conditions.

Faster R-CNN

Several training runs were performed with a ResNet backbone. A 14-epoch setup with 5-fold cross-validation and standard augmentations (random flip, photometric distortion) achieved the strongest performance at 0.200. Longer training (30 epochs) under the same setup reduced performance slightly to 0.196, possibly due to overfitting. Simpler train/validation splits yielded similar but slightly lower results (0.192), and a default configuration without augmentations performed poorly (0.130).

EfficientDet-D2

Two major configurations were tested. With a learning rate of 0.01 on a train/validation split, the model achieved 0.130. A tuned configuration with a lower learning rate (0.001), warm-up training, and weight decay significantly improved results to 0.210, the best score among the EfficientDet variants.

YOLOv5

The baseline proof-of-concept run of the default YOLOv5 model with 50 epochs and 5-fold CV reached 0.128. More advanced pipelines that combined TTA with multi-scale inputs $(640, 800, 480)$ and Mosaic/Affine augmentations gave substantial improvements. The strongest result was 0.200 on the label-4 dataset (CV, 50 epochs). Similar scores were obtained with train/validation splits (0.198). When trained on label-1, label-2, and label-3 subsets, performance declined progressively, reflecting the reduced data availability and increased label agreement requirements. Without TTA, scores dropped by around 0.01–0.02 across all splits, confirming the benefits of augmented inference.

VFNet

With learning rate 0.001 and no photometric distortion, VFNet reached 0.151 on 5-fold CV. A lower learning rate of 0.0001 combined with photometric distortion improved the score to 0.191, and the same configuration on a train/validation split gave 0.183. These results suggest that VFNet is particularly sensitive to both learning rate tuning and augmentation strategies.

To summarize, the experiments highlighted consistent trends: careful hyperparameter tuning and augmentations such as TTA or photometric distortion generally yielded notable gains, while overly long training schedules sometimes hurt generalization. Figure 5.1 provides a visual comparison of all baseline runs, grouped by model type.

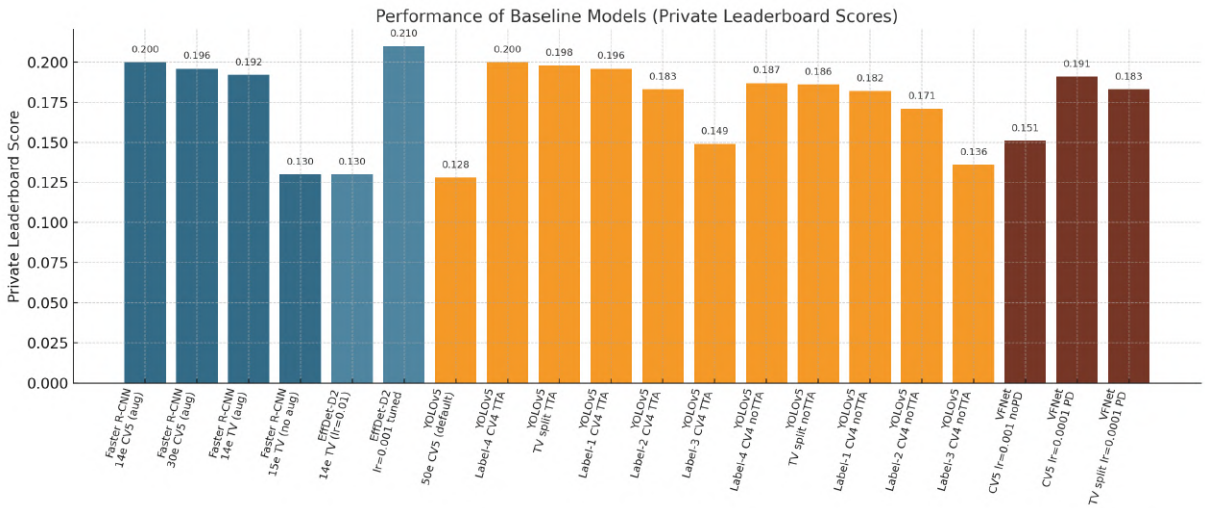


Figure 5.1: Private leaderboard scores of baseline models, grouped by architecture. Each bar corresponds to a specific training configuration.

Across all architectures, the best-performing configurations were preserved for later ensemble experiments. The configuration files are made available in the public repository to ensure transparency and reproducibility.

5.3 Insights and Validation

The baseline experiments provided several important insights into the behavior of different detection architectures on the VinBigData dataset.

One clear observation was the sensitivity of model performance to hyperparameter tuning. EfficientDet-D2, for example, performed modestly under default settings, but a carefully tuned configuration with adjusted learning rate, warm-up schedule, and weight decay produced the strongest results among all baselines. This highlights how relatively small adjustments in training parameters can make the difference between weak and competitive performance.

Augmentation strategies also played a central role. YOLOv5 showed only limited success in its default configuration, but the addition of Mosaic/Affine transformations and multi-scale test-time augmentation made it substantially more competitive. Similarly, VFNet demonstrated clear improvements when photometric distortion was introduced, suggesting that some architectures are particularly sensitive to image-level regularization techniques.

Not all attempts to improve performance were successful. Faster R-CNN, while reliable, reached its best performance relatively quickly and offered little benefit from extended training. In fact, longer training schedules appeared to reduce generalization slightly, indicating mild overfitting. This shows the importance of not simply training models longer but instead finding balanced schedules tailored to each architecture.

Overall, the baseline results confirmed that no single model configuration was sufficient to reach the levels reported by competition winners. While careful tuning and augmentation strategies could narrow the gap, the differences in performance highlighted the role of ensembling and architectural diversity in achieving state-of-the-art results. These baseline experiments therefore served a dual purpose: first, as a proof of concept to validate the correctness and reproducibility of the training pipeline, and second, as a foundation for more advanced experiments. The lessons drawn here informed the design choices in the next phase, where newer architectures and improved training strategies were explored.

Chapter 6

Results and Discussion

This chapter reports the empirical findings and interprets their implications for chest X–ray abnormality detection. Reproductions of three of the top Kaggle solutions are first presented to assess the robustness of the original ensembling strategies and to quantify the sensitivity of leaderboard scores to fusion hyperparameters. The analysis then shifts to Phase 2, where transformer–based detectors are evaluated under the same standardized pipeline to examine whether recent advances translate from natural–image benchmarks to medical imaging. Quantitative results are complemented by qualitative examples to highlight typical success and failure modes, followed by a comparative discussion that links observed performance to architectural choices, annotation noise, and computational constraints. Together, these results provide an evidence–based foundation for the conclusions and recommendations stated in the closing chapter.

6.1 Ensembling and Solutions Replication

To validate the robustness of the original competition approaches, three top Kaggle solutions (first, fourth, and tenth place) were carefully reproduced and evaluated on both the public and private leaderboards. The replication process focused not only on model configurations but also on ensembling strategies, which proved central to achieving strong leaderboard performance.

First-Place Solution

The first-place solution relied on a large ensemble of **17 models**, spanning EfficientDet-D2, YOLOv5, and Faster R-CNN with ResNet101 backbones. Replication showed how ensemble weighting and IoU thresholds influenced performance. A balanced weighting across all models produced a score of **0.241 private / 0.211 public**, while stricter ensembling strategies (precision-oriented with IoU = 0.60) reduced false positives but slightly lowered overall mAP. Recall-oriented settings (IoU = 0.45) achieved 0.237 private / 0.210 public, demonstrating the trade-off between precision and recall that ensembles must balance.

Fourth-Place Solution

The fourth-place approach centered on YOLOv5 ensembles with **12–16 models** trained on four annotation agreement levels (label-1 through label-4). Replication confirmed the

strong effect of test-time augmentation (TTA) and careful IoU tuning during Weighted Box Fusion. Interestingly, in the original description the best results were achieved by ensembling **labels 1–2–4**, whereas in our replication the highest score came from ensembling **all four label groups (1–2–3–4)** together. This full-label ensemble achieved **0.226 private / 0.199 public** at IoU = 0.40, suggesting that the additional diversity across annotations provided more benefit in this reproduced setting.

Tenth-Place Solution

The tenth-place solution combined a diverse set of **12 models**: 5 YOLOv5, 1 Faster R-CNN, and 6 VFNet. This strategy highlighted the benefit of architectural diversity, with complementary strengths stabilizing overall results. Replication showed the best outcome at IoU = 0.60, reaching **0.235 private / 0.201 public**. Both lower and higher IoU thresholds gave weaker results, underscoring the importance of tuning ensembling hyperparameters. Compared with the first-place ensemble, this setup was more compact yet still competitive, showing that balanced diversity can be more effective than scaling up sheer model counts.

Qualitative Analysis

Prediction samples from the reproduced ensembles (Figures 6.1) illustrate how these methods perform in practice. For common abnormalities such as cardiomegaly and aortic enlargement, ensembles consistently localized findings with strong agreement across models. For subtler conditions like pulmonary fibrosis and interstitial lung disease, however, predictions varied more. In some cases, ensembles successfully merged overlapping boxes into coherent regions, while in others disagreements between models led to fragmented or missed detections. These examples highlight both the robustness of ensembles in clear cases and their limitations when faced with subtle pathologies or noisy annotations.

Summary

Across all three replications, ensembles consistently outperformed single models, confirming the central role of ensembling in the competition. At the same time, the results showed that ensemble performance is highly sensitive to the choice of IoU threshold and weighting scheme. Even with careful replication, reproduced scores remained below the official leaderboard results, reflecting the difficulty of matching the exact experimental conditions and underscoring the importance of precise reproducibility in machine learning research.

Table 6.1: Best reproduced results for the three selected Kaggle solutions.

Solution	Best IoU/Setting	Score (Private / Public)
1st Place	Balanced weighting, IoU = 0.50	0.241 / 0.211
4th Place	Ensemble of labels 1–2–3–4, IoU = 0.40	0.226 / 0.199
10th Place	Diverse ensemble, IoU = 0.60	0.235 / 0.201

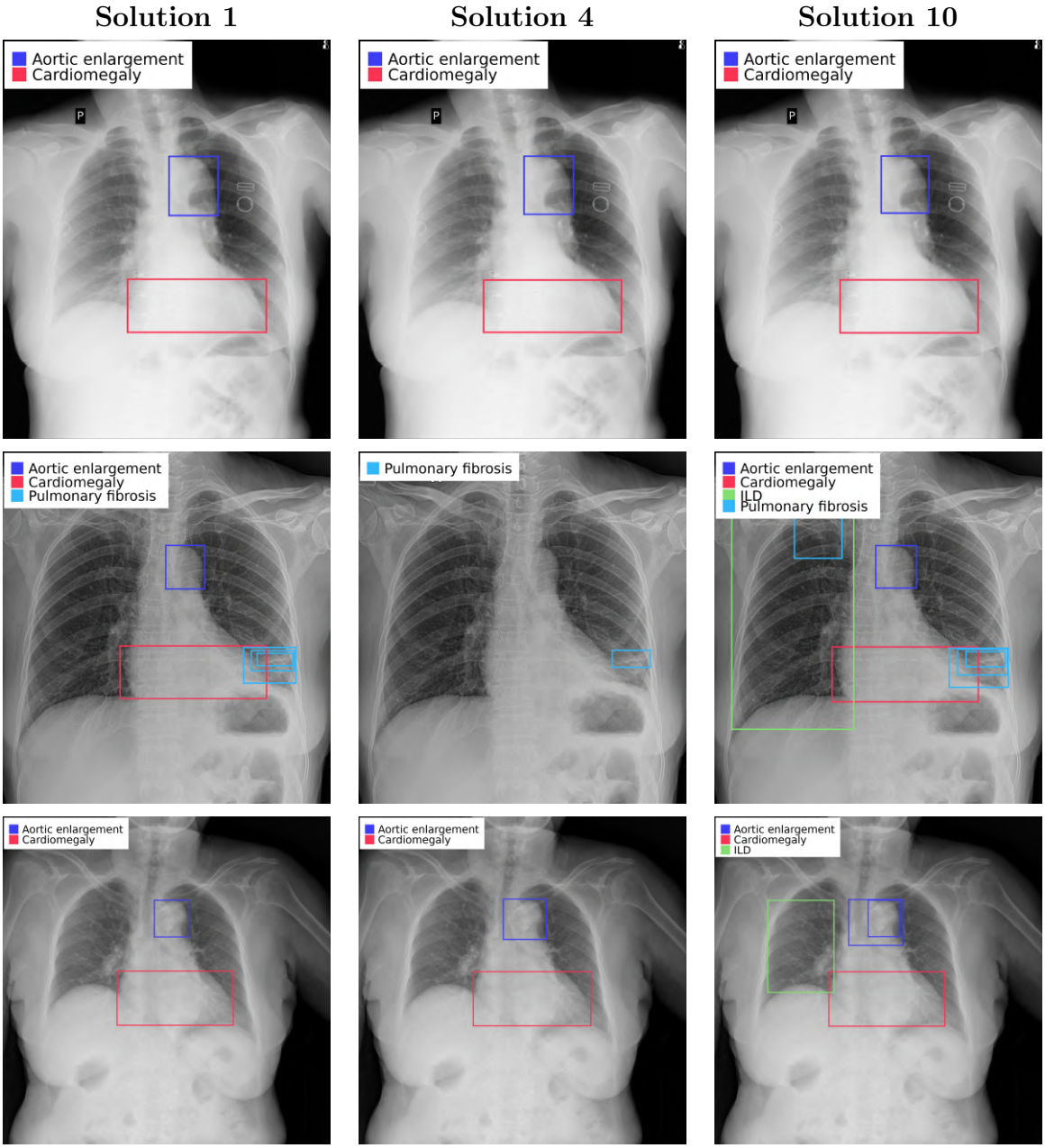


Figure 6.1: Qualitative comparison of reproduced competition solutions. Each row corresponds to one chest X-ray test image, with columns showing predictions from the 1st, 4th, and 10th place solutions.

6.2 Phase 2 Models Results

The second phase of experiments aimed to explore whether recent transformer-based detectors could outperform the convolutional baselines. In particular, two models were selected for evaluation due to their availability in MMDetection and strong reported results on the COCO benchmark: **DETR with ResNet-50 backbone** (39.9 box AP on COCO) and **Co-DETR / Co-DINO with Swin-Large backbone** (64.1 box AP on COCO). These models were chosen to test whether the significant improvements observed on natural image datasets would transfer to medical object detection tasks such as VinBigData chest X-ray abnormality detection.

Experimental setup

Both models were trained using the same preprocessing pipeline as described in Chapter 3.2, with AP50 as the principal evaluation metric. Validation was carried out every 2–5 epochs depending on training time. Unlike the baseline models, training these transformer-based detectors was much more resource-intensive: DETR required more than 100 epochs to converge, which translated into training times of over a week. Co-DETR converged more quickly in terms of epochs (stabilizing in under 15), but each epoch was substantially slower due to its heavier backbone and multi-scale attention layers. These factors limited the number of experiments that could be conducted and prevented more extensive exploration of ensembling or alternative training strategies.

Results

Table 6.2 summarizes the best results for the two transformer-based models. DETR R50 achieved up to **0.188 (private) / 0.160 (public)** in 5-fold cross-validation, while Co-DETR Swin-L reached **0.179 (private) / 0.170 (public)**. Both models delivered weaker results when trained on a single train/validation split, underscoring the stabilizing effect of cross-validation.

Table 6.2: Leaderboard results for transformer-based models in Phase 2.

Model	Private LB	Public LB
DETR R50 (CV5)	0.188	0.160
DETR R50 (Train/Val split)	0.161	0.160
Co-DETR Swin-L (CV5)	0.179	0.132
Co-DETR Swin-L (Train/Val split)	0.177	0.170

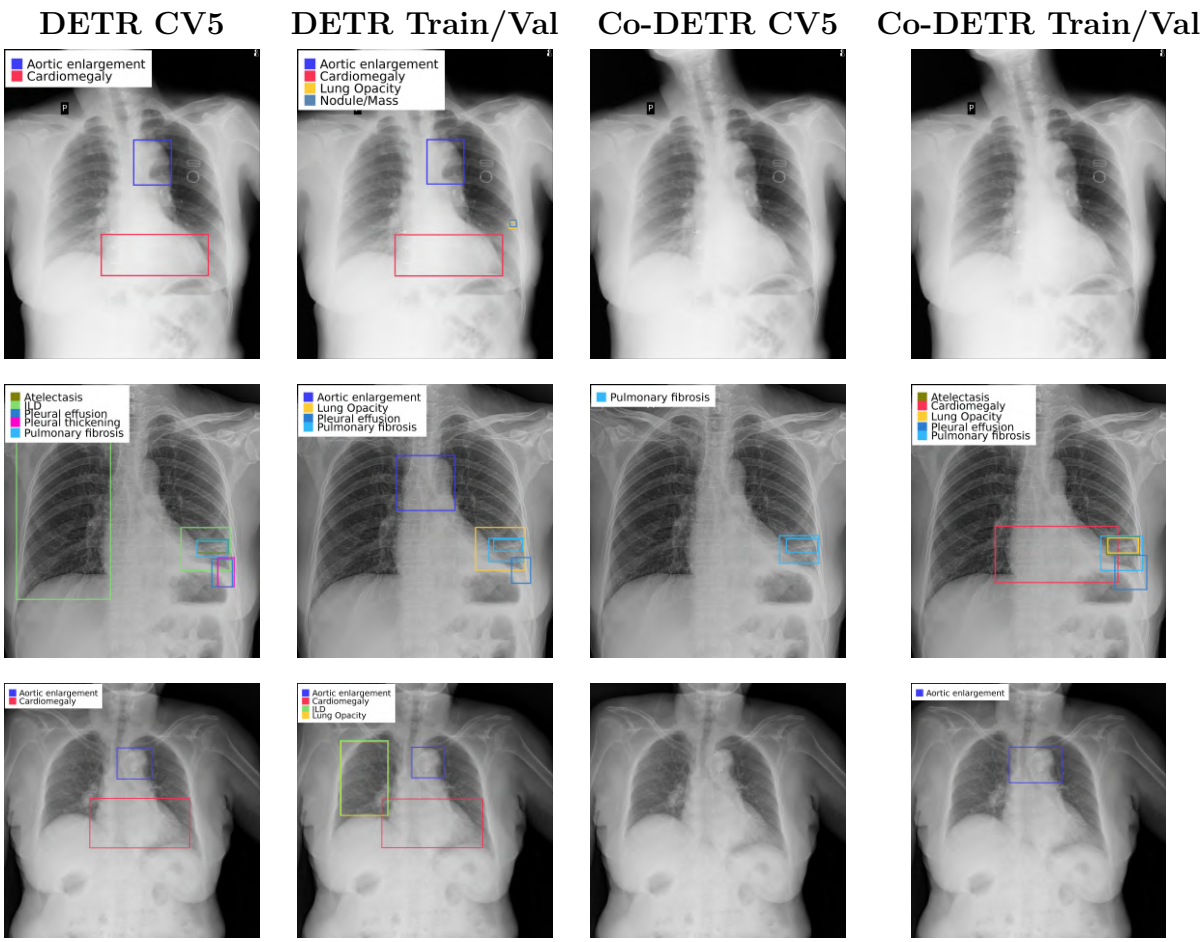


Figure 6.2: Qualitative comparison of transformer models in the second phase. Each row is one chest X-ray; columns show predictions for each model.

Despite their strong performance on COCO, both DETR R50 and Co-DETR Swin-L underperformed compared with the best convolutional baselines. For instance, the tuned EfficientDet-D2 baseline achieved 0.210 private score, while YOLOv5 with advanced augmentations reached 0.200. This contrast suggests that improvements demonstrated on natural image benchmarks do not directly translate to medical imaging datasets. Several factors may explain this discrepancy:

- **Annotation noise and label scarcity.** DETR models rely heavily on clean supervision and large datasets. In the VinBigData dataset, disagreement among radiologists and small sample sizes for rare classes may have reduced their effectiveness.
- **Training cost.** Both DETR and Co-DETR required much longer training schedules than CNN-based detectors. With limited resources, it was not possible to fully explore variations, ensembles, or deeper fine-tuning.
- **Domain gap.** Transformers were designed for general-purpose object detection, but the low-contrast and highly localized abnormalities in chest X-rays present different challenges from COCO-style natural images.

In summary, while the transformer-based models validated the correctness of the experimental pipeline and offered valuable insights, they did not surpass the best convolutional baselines in this project. This highlights the importance of careful benchmarking

in medical AI, where architectural advances may not always provide the same benefits observed in mainstream computer vision.

6.3 Additional Discussion

The second phase of this project highlighted several important challenges that help explain why the transformer-based models underperformed compared to expectations from their COCO benchmark results. The first challenge is the **lack of proper fine-tuning under time constraints**. Models such as DETR are well documented to require long training schedules, often exceeding 300 epochs, to achieve convergence on large-scale datasets. In this project, the DETR R50 model only began to stabilize after more than 100 epochs, which already required over a week of continuous training on two Tesla V100 GPUs. Due to this extended runtime, it was not feasible to perform systematic hyperparameter searches or to investigate advanced training strategies such as learning rate warm-up schedules, multi-scale augmentation, or knowledge distillation. Consequently, the reported DETR results represent only a partially optimized baseline rather than the full potential of the architecture.

For Co-DETR, convergence occurred earlier, with useful results already visible by the 15th epoch. However, each training epoch was significantly slower than CNN-based approaches, which again limited the opportunity for multiple experimental runs. The resource-intensive nature of these transformer architectures underscores a broader issue in medical imaging research: even when advanced models are available in open-source frameworks, their computational cost can make them inaccessible for thorough experimentation within the constraints of an academic thesis project. With more time and resources, fine-tuning strategies such as longer schedules, layer-wise learning rate decay, or pretraining on domain-specific chest X-ray datasets could significantly improve the performance of these models.

A second, and equally important, challenge lies in the **quality and consistency of annotations** in the VinBigData dataset. While the dataset is unique in its scale and detail, it was annotated by multiple radiologists, and differences in their interpretations are evident. Although preprocessing strategies such as Weighted Boxes Fusion were applied to reconcile bounding box placements, some disagreements go beyond geometry and extend to *semantic labeling*. In certain cases, two annotators marked almost identical regions but assigned them to different pathologies, such as interstitial lung disease versus pulmonary fibrosis. Figure 6.3 demonstrates such an example. These discrepancies are more than small labeling errors—they reflect the inherent diagnostic uncertainty of chest radiography, where different experts may reasonably disagree. From a machine learning perspective, however, such noise reduces the clarity of the learning signal and places an upper bound on the performance achievable by any detector.

Together, these factors suggest that the relatively modest performance of transformer-based models in this project should not be seen as evidence of their ineffectiveness. Rather, it reflects two limiting conditions: the **practical constraints of training time and compute resources**, and the **dataset-specific issue of annotator variability**. Addressing both would require future work: investing in longer, better-resourced training schedules for transformer architectures, and exploring annotation harmonization strategies such as consensus labeling or uncertainty-aware loss functions.

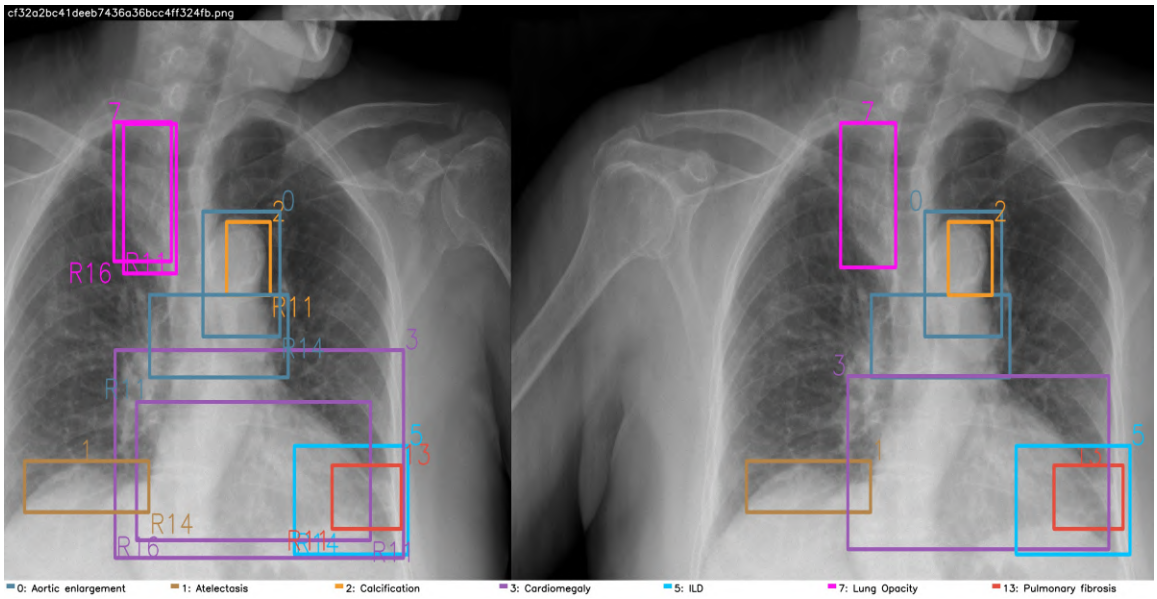


Figure 6.3: Example of annotator disagreement. Two radiologists marked nearly the same region but assigned different diagnoses (*interstitial lung disease* vs. *pulmonary fibrosis*). Such discrepancies highlight the inherent subjectivity of chest X-ray interpretation and introduce noise into the training labels.

Table 6.3: Comparison of best-performing models from Phase 1 (CNN-based) and Phase 2 (transformer-based). Scores are reported on the Kaggle VinBigData Chest X-ray competition leaderboards.

Phase	**Model**	**Private LB (PR)**	**Public LB (PB)**
Phase 1	EfficientDet-D2 (best setting)	0.210	–
Phase 1	YOLOv5 (Label-4, CV4 + TTA)	0.200	–
Phase 1	VFNet (CV5, tuned)	0.191	–
Phase 1	Faster R-CNN (30 epochs CV5)	0.196	–
Phase 2	DETR R50 (CV5)	0.188	0.160
Phase 2	DETR R50 (Train/Val split)	0.161	0.160
Phase 2	Co-DETR Swin-L (Train/Val)	0.177	0.170
Phase 2	Co-DETR Swin-L (CV5)	0.179	0.132

Chapter 7

Conclusion and Future Work

7.1 Summary of Findings

This thesis set out to reproduce and extend leading approaches from the VinBigData Chest X-ray Abnormalities Detection challenge. A central finding is the importance of dataset handling when working with medical images, particularly in contexts where even the provided ground truths carry a degree of noise. The annotation process itself proved to be a limiting factor: radiologists often disagreed on the location or extent of abnormalities, and this disagreement introduced uncertainty into the training process. WBF was employed to mitigate this effect, but residual variability remained.

The replication of top Kaggle solutions demonstrated another key challenge: reproducibility in data science competitions. Despite faithfully reconstructing model architectures and training strategies, the reproduced solutions consistently fell short of the original leaderboard results. This suggests that undocumented preprocessing details or missing hyperparameters played a significant role in shaping the original outcomes. Such gaps underline the broader reproducibility problem in this field, where seemingly minor details in configuration or pipeline design can have major effects on performance.

In terms of model performance, the experiments confirmed that convolutional architectures such as EfficientDet-D2 and YOLOv5 provided the strongest and most reliable baselines within the constraints of this project. These models offered stable training dynamics and competitive private leaderboard scores. By contrast, the transformer-based models explored in the second phase—DETR and Co-DETR—were considerably more demanding in terms of training resources and convergence time. Although they showed promise, particularly Co-DETR on train/validation splits, the limited fine-tuning time prevented them from achieving their full potential.

Overall, this work highlights two broad findings. First, the careful handling of annotations and preprocessing is at least as important as model choice when tackling medical object detection. Second, reproducibility remains a significant barrier: without fully documented experimental pipelines, even strong attempts at replication may fail to match reported results.

7.2 Contributions of The Thesis

This thesis makes several contributions to the study of automated chest X-ray abnormality detection. The first contribution is the systematic **reproduction of three top-performing Kaggle solutions** from the VinBigData competition. By reconstructing their pipelines in modern frameworks such as MMDetection, MMYOLO, and EfficientDet-PyTorch, this work not only provides transparency but also exposes the reproducibility gap between public competition reports and independently verified experiments.

A second contribution lies in the creation of a **unified experimental framework**. All dataset preprocessing, model training, and evaluation steps were carefully documented and modularized, with a strong emphasis on compatibility with the COCO format. This ensured that experiments were consistent across different model families, enabling fair comparisons between one-stage, two-stage, and transformer-based architectures.

Third, the thesis contributes a **baseline benchmark of state-of-the-art object detectors** on the VinBigData chest X-ray dataset. The experiments provide clear evidence of the strengths of convolutional models like EfficientDet and YOLOv5 under controlled conditions, while also showing the challenges of training transformer-based models such as DETR and Co-DETR in practice. The benchmark results, alongside the detailed descriptions of training strategies, offer a reliable reference point for future work.

Finally, the thesis provides an **open-source resource for the research community**. The full repository—including preprocessing scripts, configuration files, and trained model setups—will be made publicly available. This commitment to open science not only allows other researchers to reproduce the results presented here but also to extend them, whether by fine-tuning transformers more thoroughly, exploring new ensembling strategies, or addressing annotation noise in different ways.

7.3 Recommendations for Future Research

Building on the findings of this thesis, several promising directions emerge for future research. One important area concerns the **data preprocessing pipeline**. The experiments here demonstrated that annotation noise is a critical bottleneck, limiting model performance even when strong architectures are used. Future work could adopt a more **data-driven approach** to preprocessing, systematically exploring how new methods for label denoising, annotator consensus modeling, or semi-supervised correction affect downstream detection performance.

A second avenue relates to **post-processing strategies**. In this thesis, WBF was employed as the main method of consolidating predictions. While effective, WBF is not the only option. Alternative strategies such as NMS, Soft-NMS [57], or learning-based ensembling approaches could be investigated. These methods are typically less resource-intensive than training additional models, yet they may substantially improve performance if tuned effectively. Exploring the trade-offs between different post-processing schemes could yield valuable insights into how to maximize performance under real-world constraints.

A further recommendation is to extend the exploration of **transformer-based object detectors**. The limited experiments with DETR and Co-DETR suggest that transformers have potential but require extensive fine-tuning and computational resources to realize their advantages. Future studies could investigate newer transformer architectures that address known limitations, such as Deformable DETR [26], Conditional DETR [28], or DINO [56]. Evaluating these models under carefully designed experimental protocols could help clarify whether transformers can outperform convolutional approaches in medical imaging domains.

Finally, there is scope for combining these directions into a holistic approach. For example, integrating advanced preprocessing to reduce annotation noise, experimenting with post-processing techniques beyond WBF, and applying next-generation transformers may together establish a new benchmark for chest X-ray abnormality detection. Such an integrated perspective would also contribute to the broader challenge of improving reproducibility and generalization in medical AI.

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